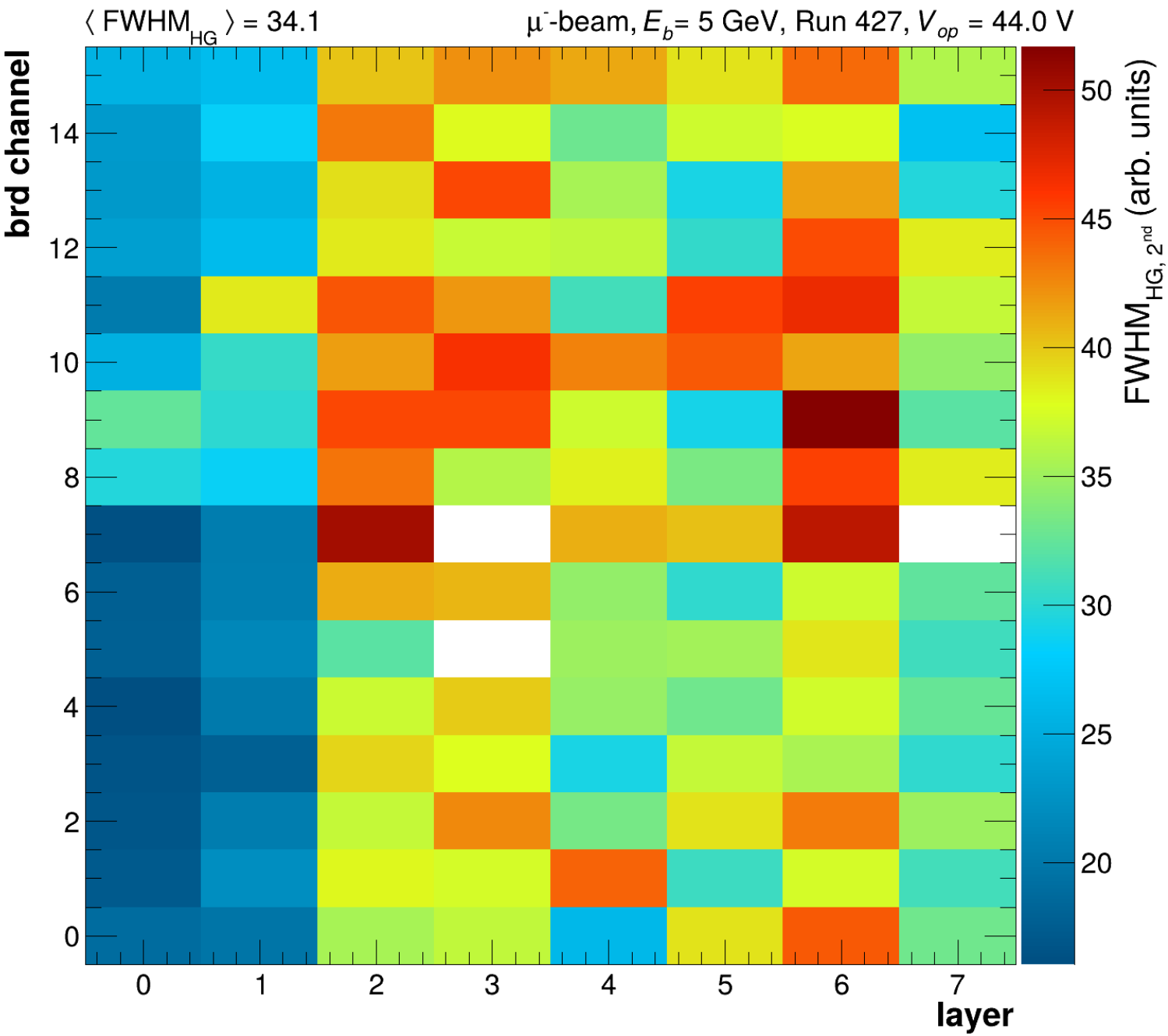
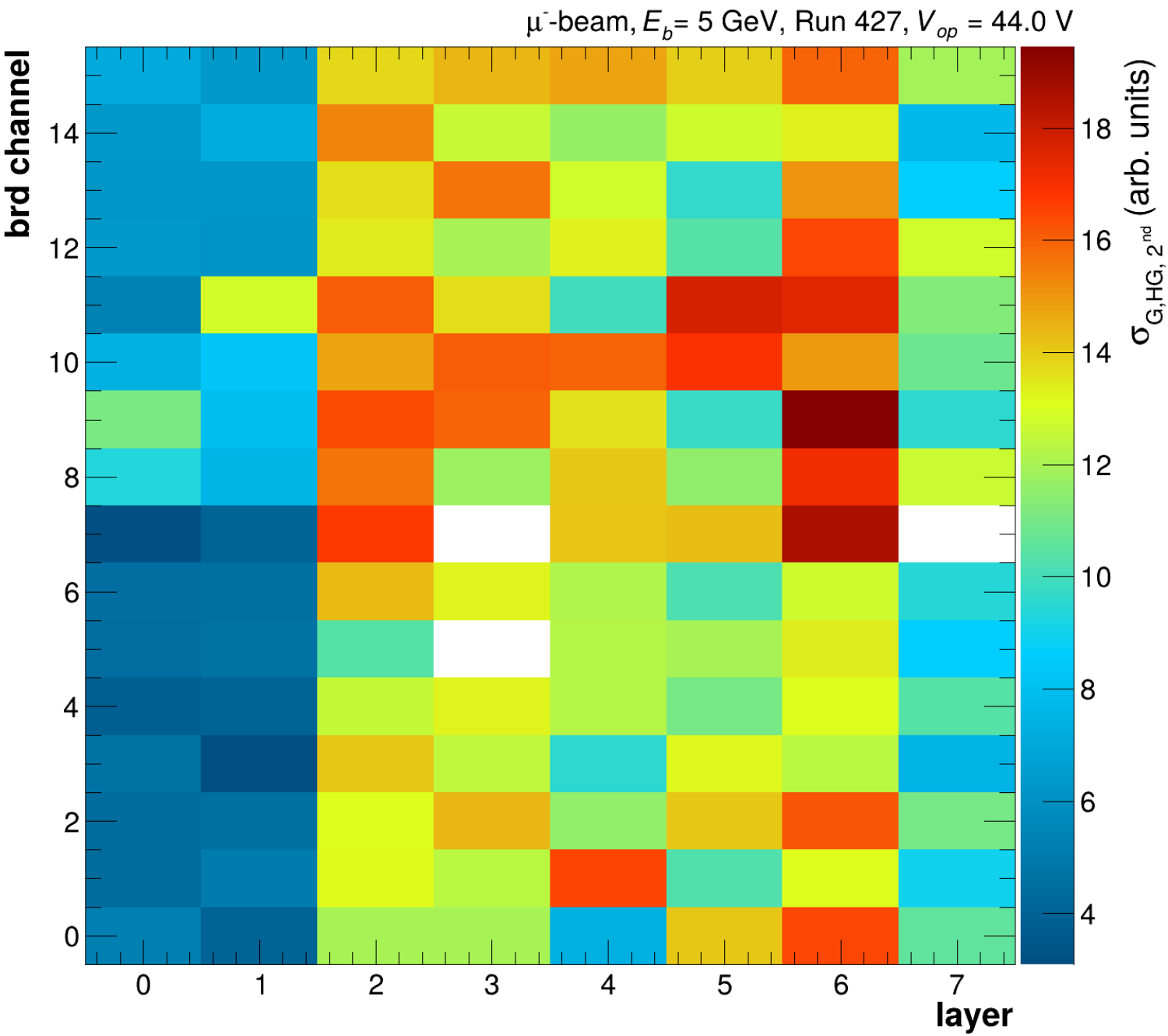
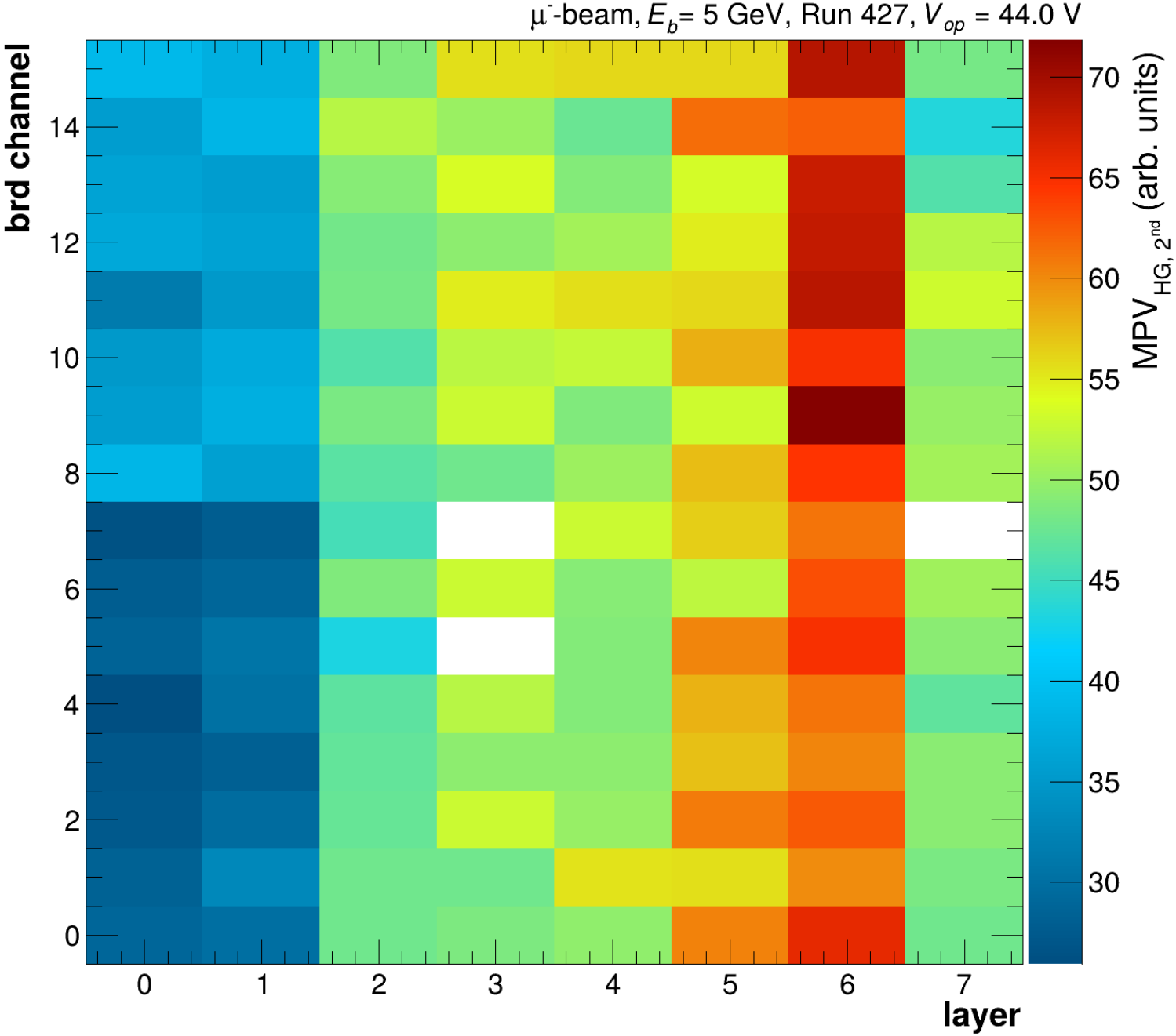
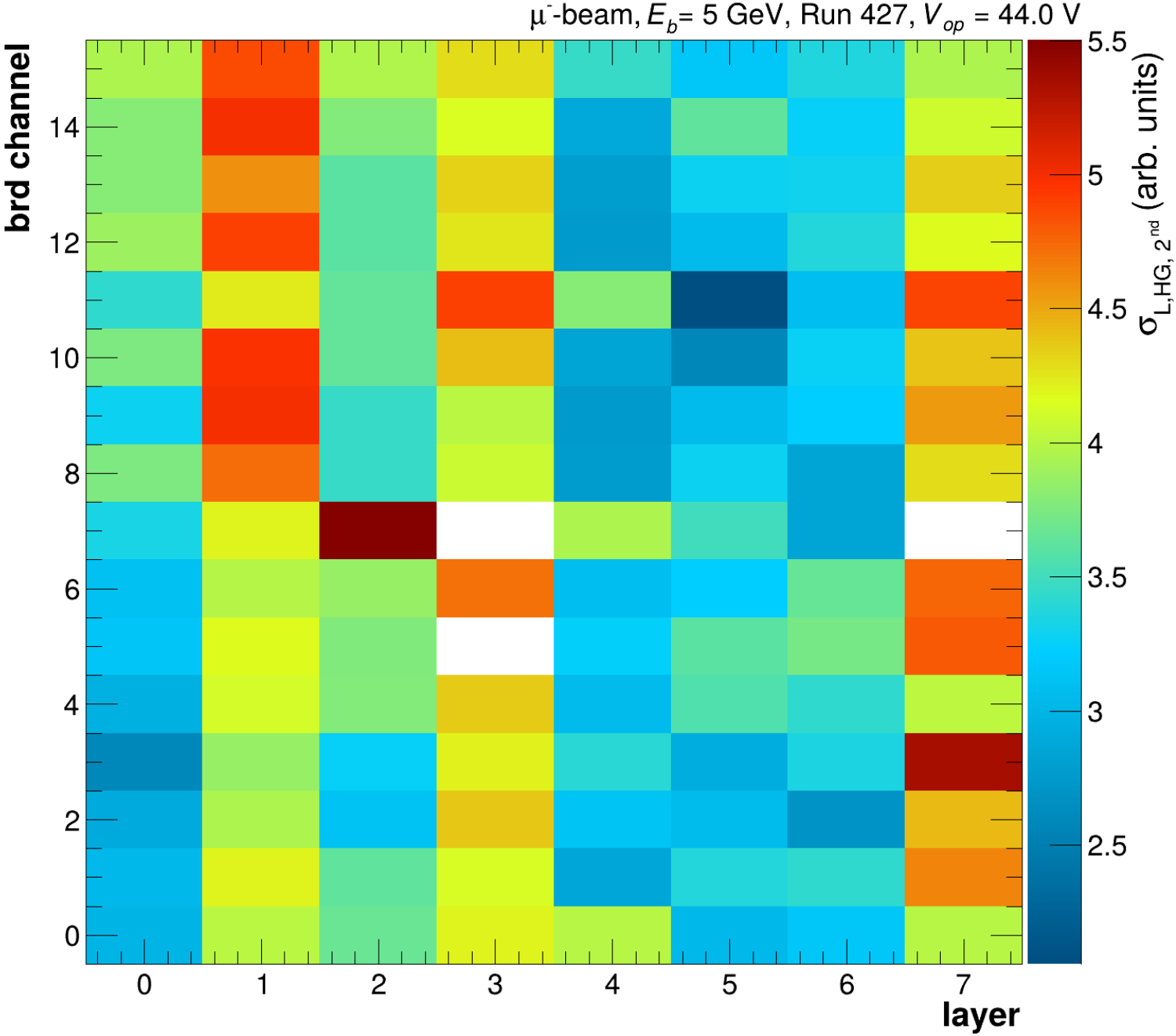
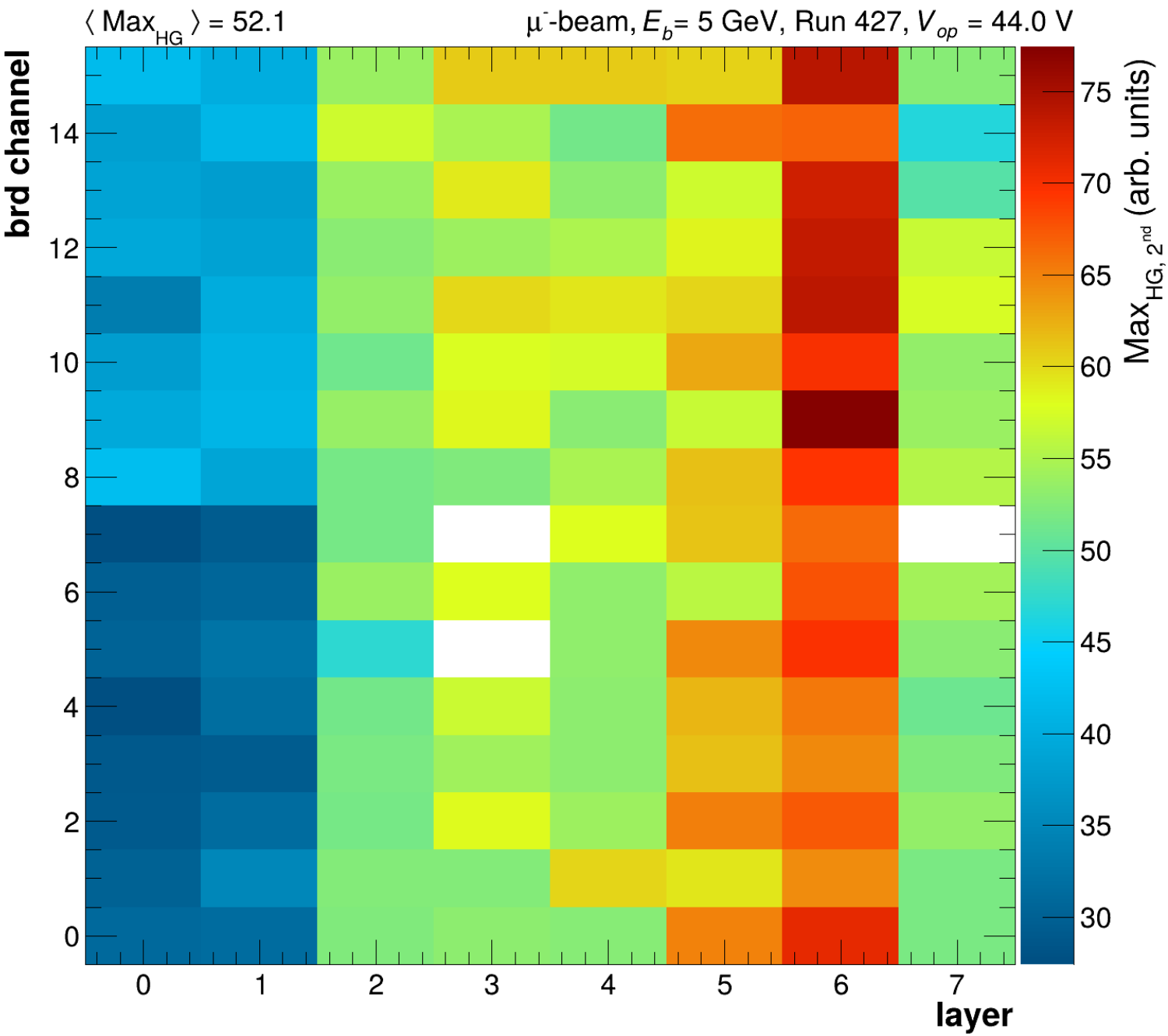
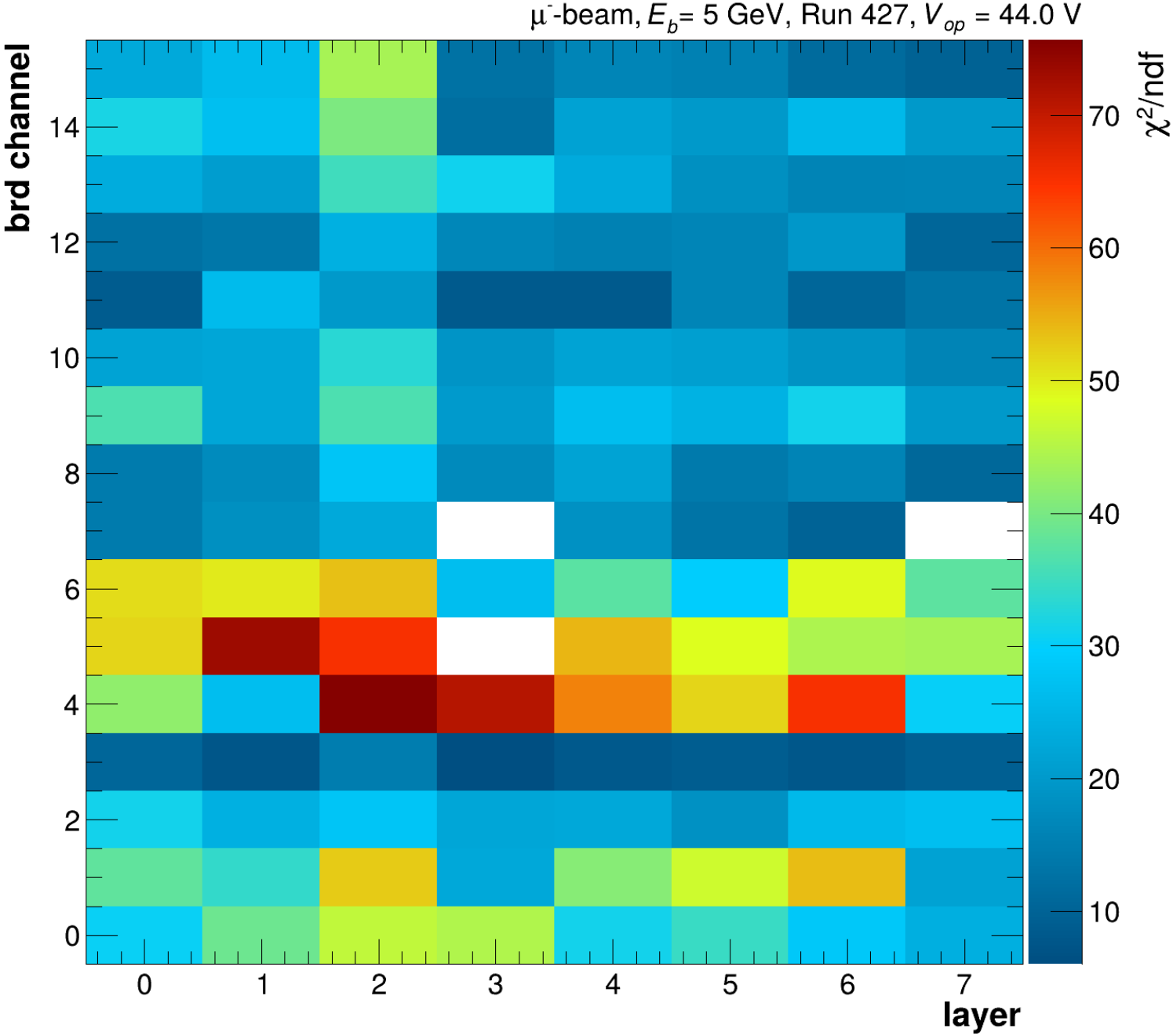

⟨ FWHM$_{HG}$ ⟩ = 34.1
μ$^-$-beam, E_b= 5 GeV, Run 427, V_{op} = 44.0 V
brd channel
layer
FWHM$_{HG,\ 2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
20
25
30
35
40
45
50

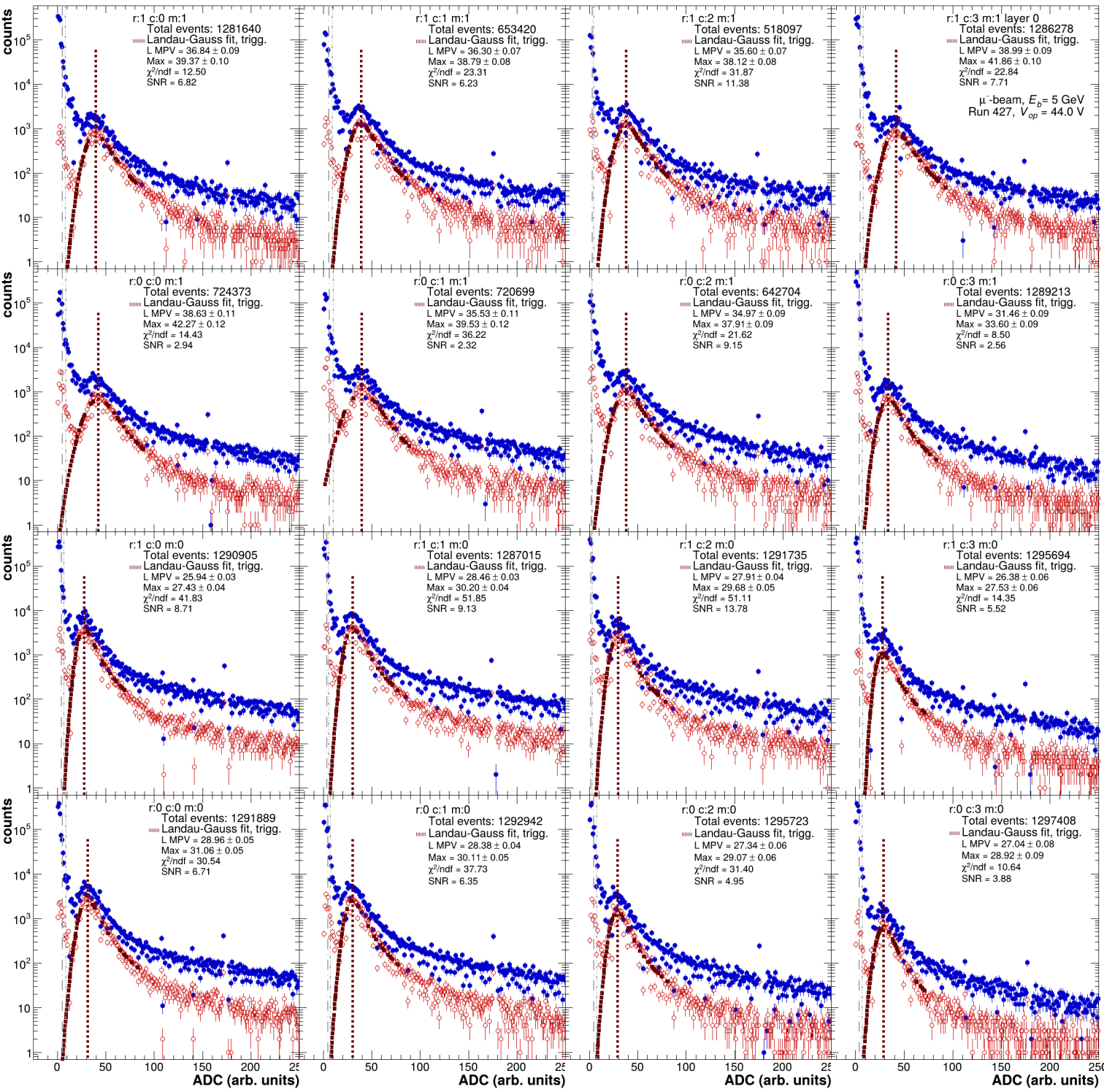

μ⁻-beam, E_b= 5 GeV, Run 427, V_{op} = 44.0 V
brd channel
layer
$\sigma_{G,HG,2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
4
6
8
10
12
14
16
18

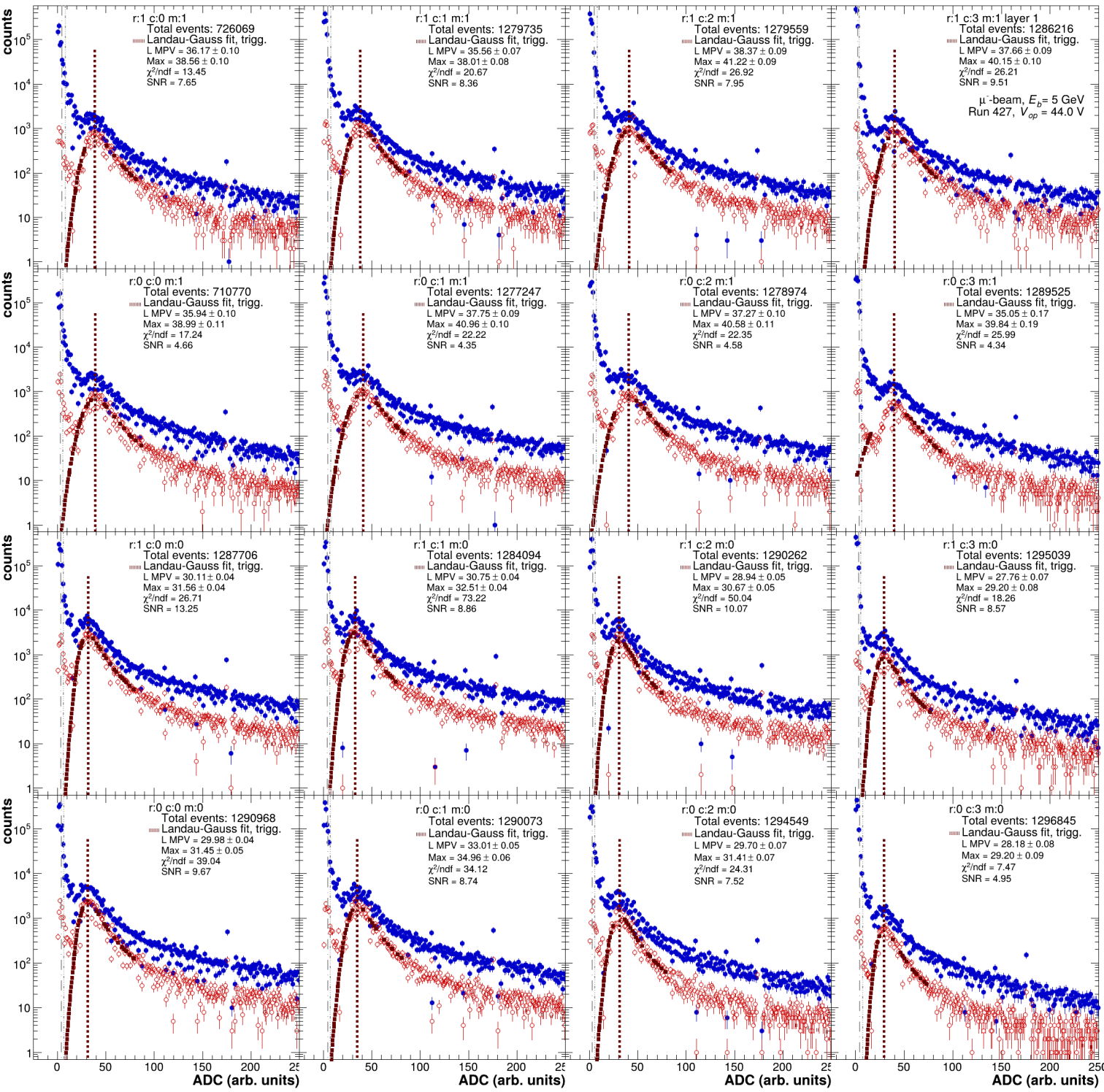

μ⁻-beam, E_b= 5 GeV, Run 427, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
MPV$_{HG,\,2^{nd}}$ (arb. units)
30
35
40
45
50
55
60
65
70

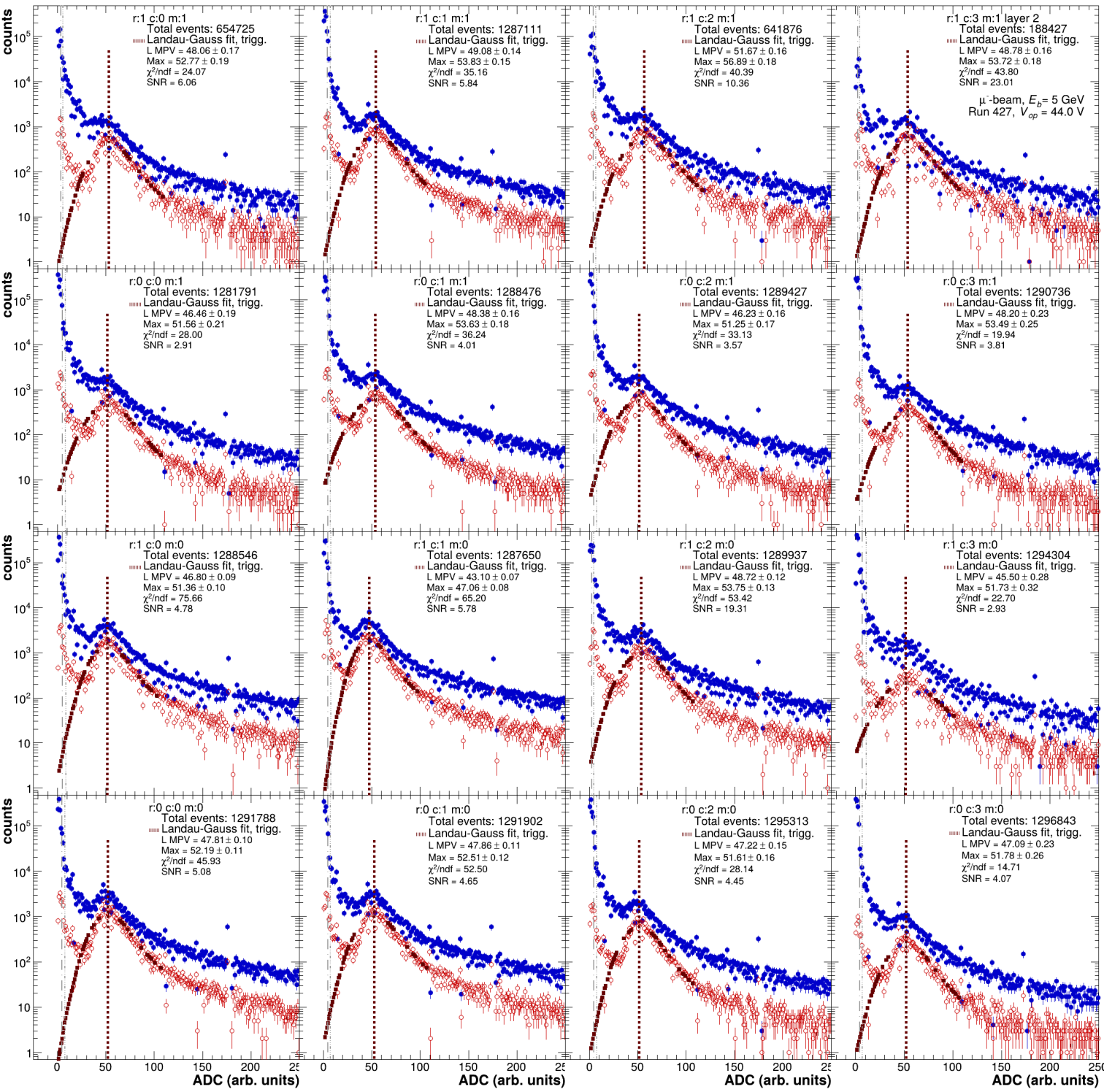

μ⁻-beam, E_b= 5 GeV, Run 427, V_{op} = 44.0 V
brd channel
layer
$\sigma_{L,HG,2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
2.5
3
3.5
4
4.5
5
5.5

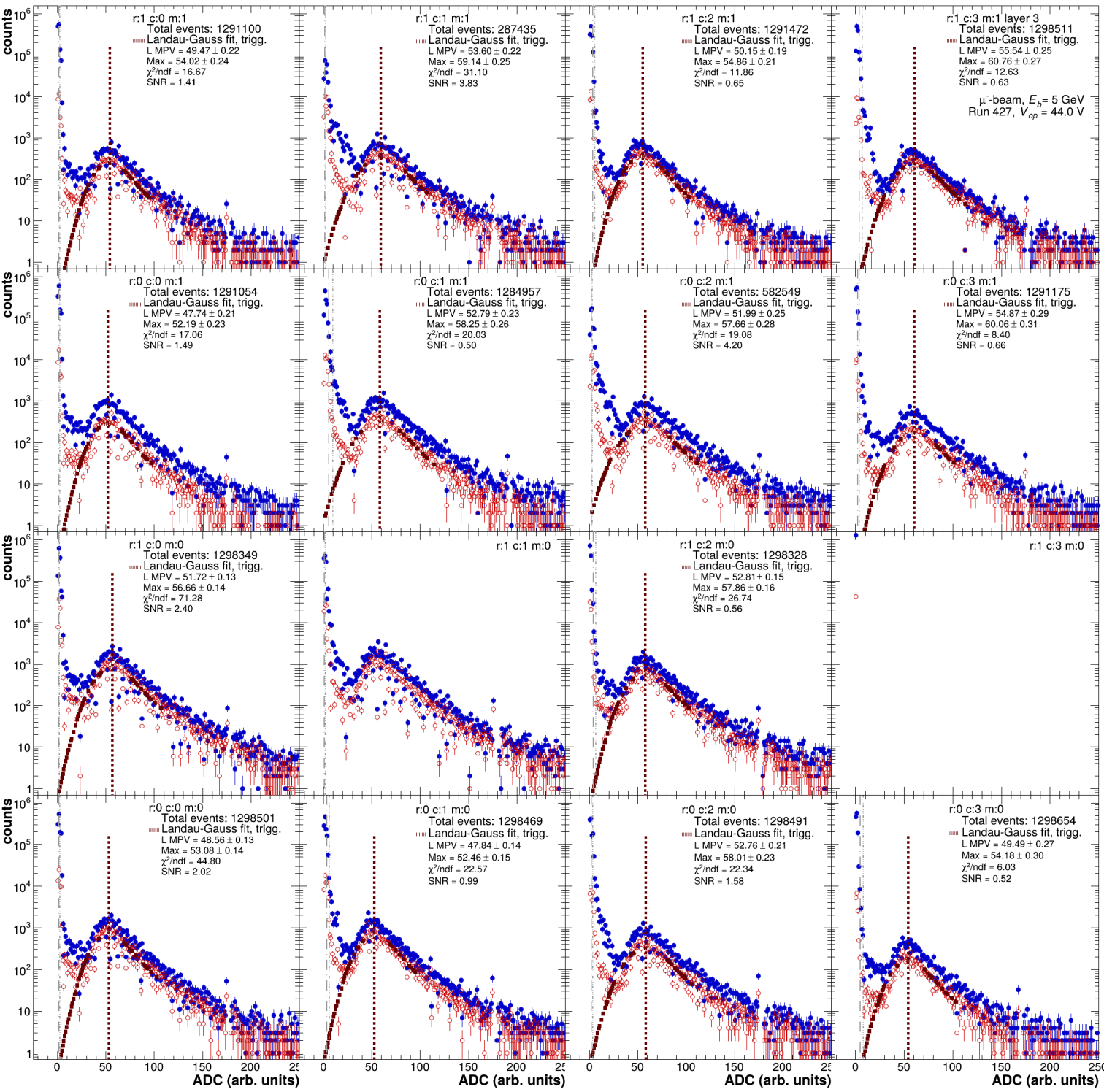

⟨ Max_HG ⟩ = 52.1
μ⁻-beam, E_b = 5 GeV, Run 427, V_op = 44.0 V
brd channel
layer
Max_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
30
35
40
45
50
55
60
65
70
75

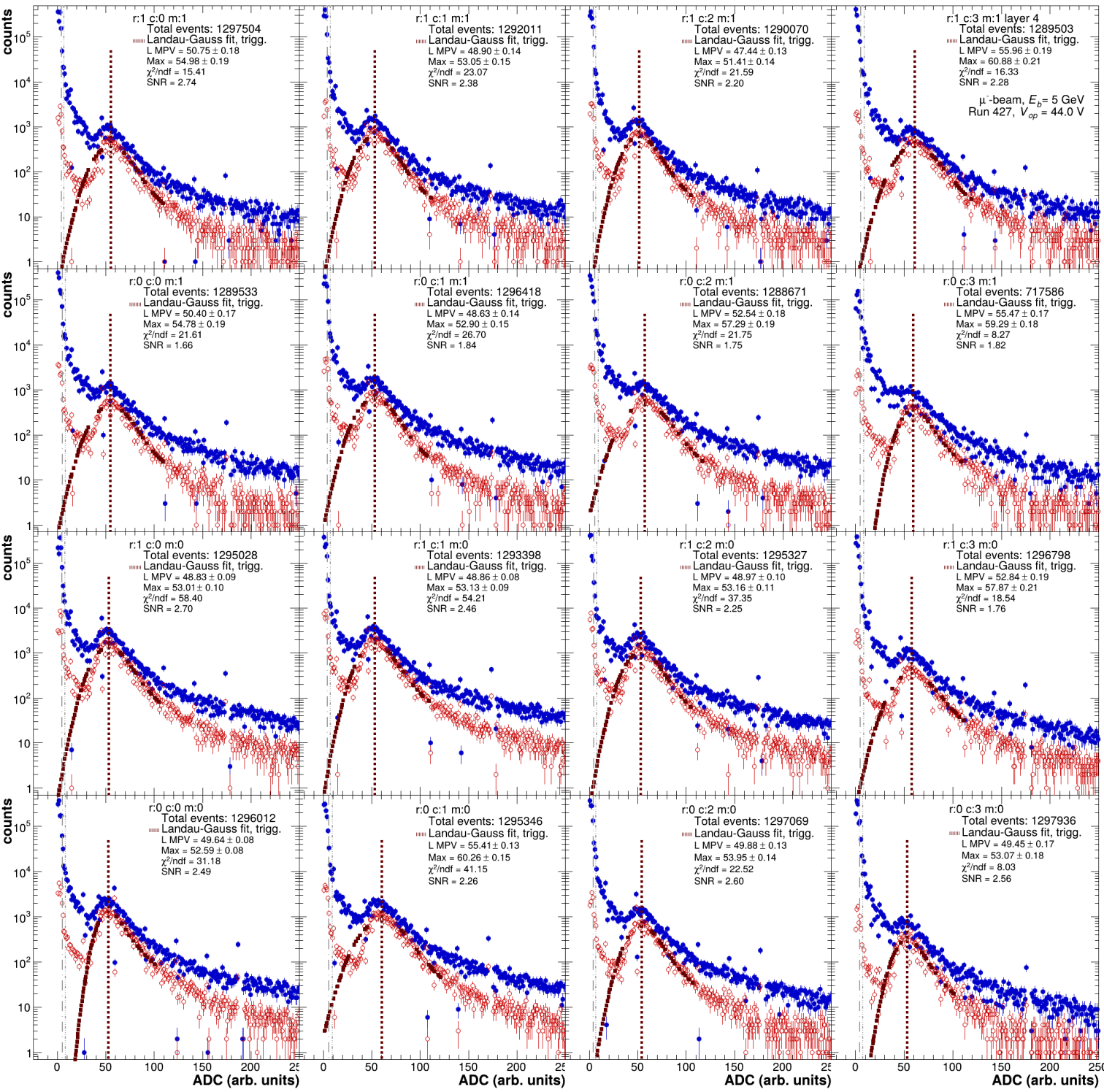

μ⁻-beam, E_b= 5 GeV, Run 427, V_{op} = 44.0 V
brd channel
layer
χ^2/ndf
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
10
20
30
40
50
60
70

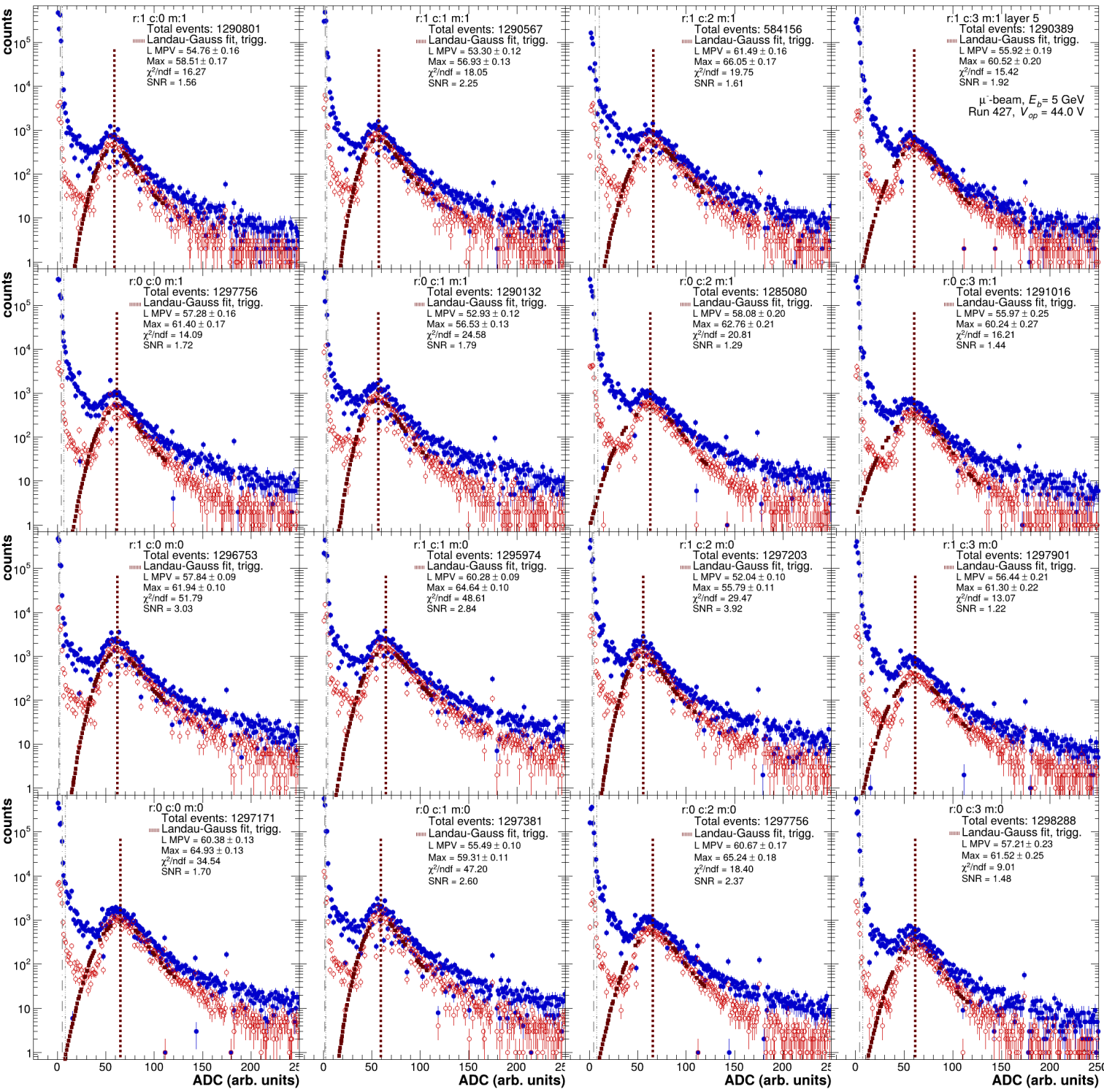

r:1 c:0 m:1
Total events: 1281640
Landau-Gauss fit, trigg.
L MPV = 36.84 ± 0.09
Max = 39.37 ± 0.10
χ²/ndf = 12.50
SNR = 6.82
r:1 c:1 m:1
Total events: 653420
Landau-Gauss fit, trigg.
L MPV = 36.30 ± 0.07
Max = 38.79 ± 0.08
χ²/ndf = 23.31
SNR = 6.23
r:1 c:2 m:1
Total events: 518097
Landau-Gauss fit, trigg.
L MPV = 35.60 ± 0.07
Max = 38.12 ± 0.08
χ²/ndf = 31.87
SNR = 11.38
r:1 c:3 m:1 layer 0
Total events: 1286278
Landau-Gauss fit, trigg.
L MPV = 38.99 ± 0.09
Max = 41.86 ± 0.10
χ²/ndf = 22.84
SNR = 7.71
μ⁻-beam, E_b = 5 GeV
Run 427, V_op = 44.0 V
r:0 c:0 m:1
Total events: 724373
Landau-Gauss fit, trigg.
L MPV = 38.63 ± 0.11
Max = 42.27 ± 0.12
χ²/ndf = 14.43
SNR = 2.94
r:0 c:1 m:1
Total events: 720699
Landau-Gauss fit, trigg.
L MPV = 35.53 ± 0.11
Max = 39.53 ± 0.12
χ²/ndf = 36.22
SNR = 2.32
r:0 c:2 m:1
Total events: 642704
Landau-Gauss fit, trigg.
L MPV = 34.97 ± 0.09
Max = 37.91 ± 0.09
χ²/ndf = 21.62
SNR = 9.15
r:0 c:3 m:1
Total events: 1289213
Landau-Gauss fit, trigg.
L MPV = 31.46 ± 0.09
Max = 33.60 ± 0.09
χ²/ndf = 8.50
SNR = 2.56
r:1 c:0 m:0
Total events: 1290905
Landau-Gauss fit, trigg.
L MPV = 25.94 ± 0.03
Max = 27.43 ± 0.04
χ²/ndf = 41.83
SNR = 8.71
r:1 c:1 m:0
Total events: 1287015
Landau-Gauss fit, trigg.
L MPV = 28.46 ± 0.03
Max = 30.20 ± 0.04
χ²/ndf = 51.85
SNR = 9.13
r:1 c:2 m:0
Total events: 1291735
Landau-Gauss fit, trigg.
L MPV = 27.91 ± 0.04
Max = 29.68 ± 0.05
χ²/ndf = 51.11
SNR = 13.78
r:1 c:3 m:0
Total events: 1295694
Landau-Gauss fit, trigg.
L MPV = 26.38 ± 0.06
Max = 27.53 ± 0.06
χ²/ndf = 14.35
SNR = 5.52
r:0 c:0 m:0
Total events: 1291889
Landau-Gauss fit, trigg.
L MPV = 28.96 ± 0.05
Max = 31.06 ± 0.05
χ²/ndf = 30.54
SNR = 6.71
r:0 c:1 m:0
Total events: 1292942
Landau-Gauss fit, trigg.
L MPV = 28.38 ± 0.04
Max = 30.11 ± 0.05
χ²/ndf = 37.73
SNR = 6.35
r:0 c:2 m:0
Total events: 1295723
Landau-Gauss fit, trigg.
L MPV = 27.34 ± 0.06
Max = 29.07 ± 0.06
χ²/ndf = 31.40
SNR = 4.95
r:0 c:3 m:0
Total events: 1297408
Landau-Gauss fit, trigg.
L MPV = 27.04 ± 0.08
Max = 28.92 ± 0.09
χ²/ndf = 10.64
SNR = 3.88
counts
ADC (arb. units)

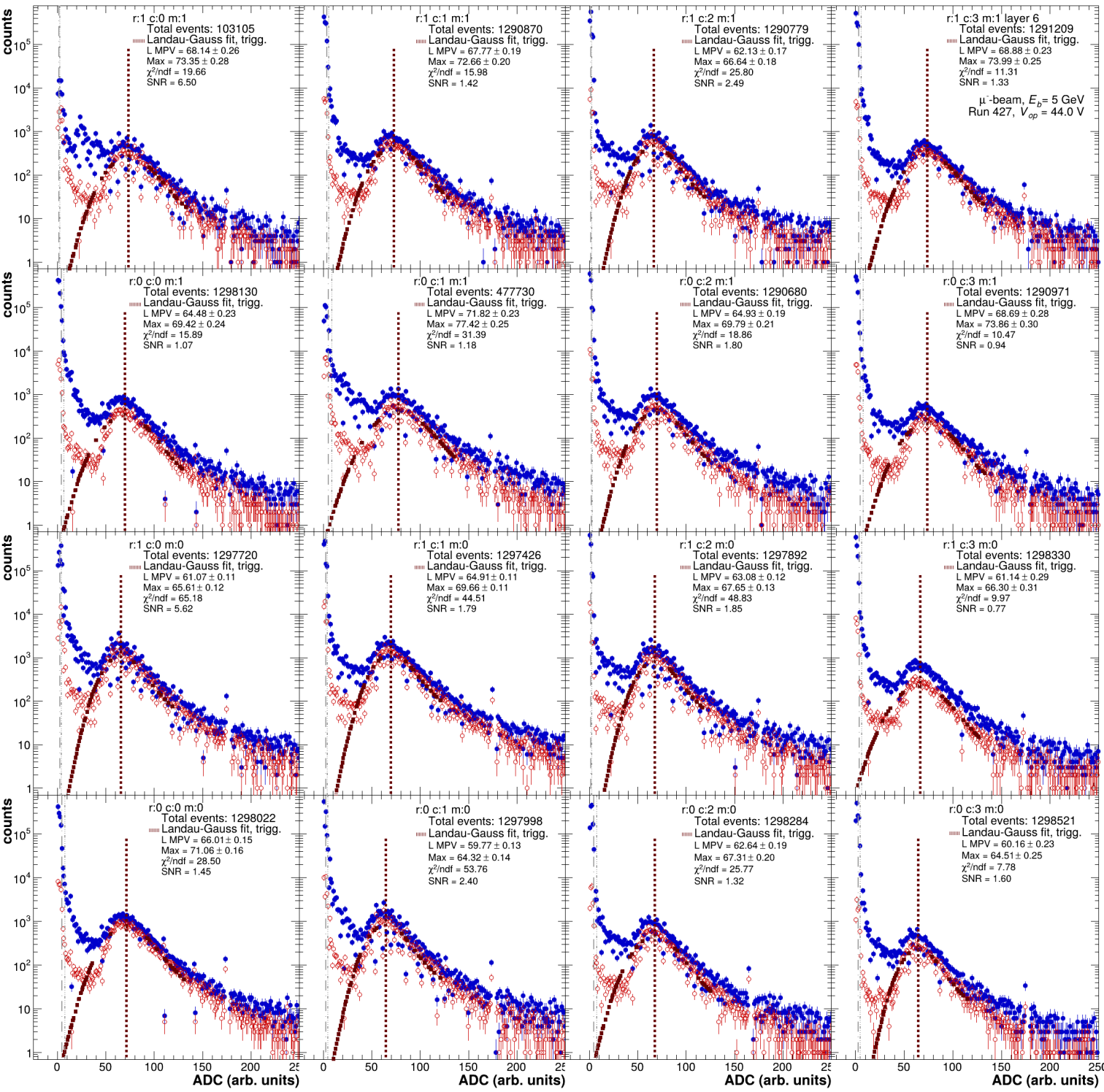

r:1 c:0 m:1
Total events: 726069
Landau-Gauss fit, trigg.
L MPV = 36.17 ± 0.10
Max = 38.56 ± 0.10
χ²/ndf = 13.45
SNR = 7.65
r:1 c:1 m:1
Total events: 1279735
Landau-Gauss fit, trigg.
L MPV = 35.56 ± 0.07
Max = 38.01 ± 0.08
χ²/ndf = 20.67
SNR = 8.36
r:1 c:2 m:1
Total events: 1279559
Landau-Gauss fit, trigg.
L MPV = 38.37 ± 0.09
Max = 41.22 ± 0.09
χ²/ndf = 26.92
SNR = 7.95
r:1 c:3 m:1 layer 1
Total events: 1286216
Landau-Gauss fit, trigg.
L MPV = 37.66 ± 0.09
Max = 40.15 ± 0.10
χ²/ndf = 26.21
SNR = 9.51
μ-beam, E_b = 5 GeV
Run 427, V_op = 44.0 V
r:0 c:0 m:1
Total events: 710770
Landau-Gauss fit, trigg.
L MPV = 35.94 ± 0.10
Max = 38.99 ± 0.11
χ²/ndf = 17.24
SNR = 4.66
r:0 c:1 m:1
Total events: 1277247
Landau-Gauss fit, trigg.
L MPV = 37.75 ± 0.09
Max = 40.96 ± 0.10
χ²/ndf = 22.22
SNR = 4.35
r:0 c:2 m:1
Total events: 1278974
Landau-Gauss fit, trigg.
L MPV = 37.27 ± 0.10
Max = 40.58 ± 0.11
χ²/ndf = 22.35
SNR = 4.58
r:0 c:3 m:1
Total events: 1289525
Landau-Gauss fit, trigg.
L MPV = 35.05 ± 0.17
Max = 39.84 ± 0.19
χ²/ndf = 25.99
SNR = 4.34
r:1 c:0 m:0
Total events: 1287706
Landau-Gauss fit, trigg.
L MPV = 30.11 ± 0.04
Max = 31.56 ± 0.04
χ²/ndf = 26.71
SNR = 13.25
r:1 c:1 m:0
Total events: 1284094
Landau-Gauss fit, trigg.
L MPV = 30.75 ± 0.04
Max = 32.51 ± 0.04
χ²/ndf = 73.22
SNR = 8.86
r:1 c:2 m:0
Total events: 1290262
Landau-Gauss fit, trigg.
L MPV = 28.94 ± 0.05
Max = 30.67 ± 0.05
χ²/ndf = 50.04
SNR = 10.07
r:1 c:3 m:0
Total events: 1295039
Landau-Gauss fit, trigg.
L MPV = 27.76 ± 0.07
Max = 29.20 ± 0.08
χ²/ndf = 18.26
SNR = 8.57
r:0 c:0 m:0
Total events: 1290968
Landau-Gauss fit, trigg.
L MPV = 29.98 ± 0.04
Max = 31.45 ± 0.05
χ²/ndf = 39.04
SNR = 9.67
r:0 c:1 m:0
Total events: 1290073
Landau-Gauss fit, trigg.
L MPV = 33.01 ± 0.05
Max = 34.96 ± 0.06
χ²/ndf = 34.12
SNR = 8.74
r:0 c:2 m:0
Total events: 1294549
Landau-Gauss fit, trigg.
L MPV = 29.70 ± 0.07
Max = 31.41 ± 0.07
χ²/ndf = 24.31
SNR = 7.52
r:0 c:3 m:0
Total events: 1296845
Landau-Gauss fit, trigg.
L MPV = 28.18 ± 0.08
Max = 29.20 ± 0.09
χ²/ndf = 7.47
SNR = 4.95
counts
ADC (arb. units)

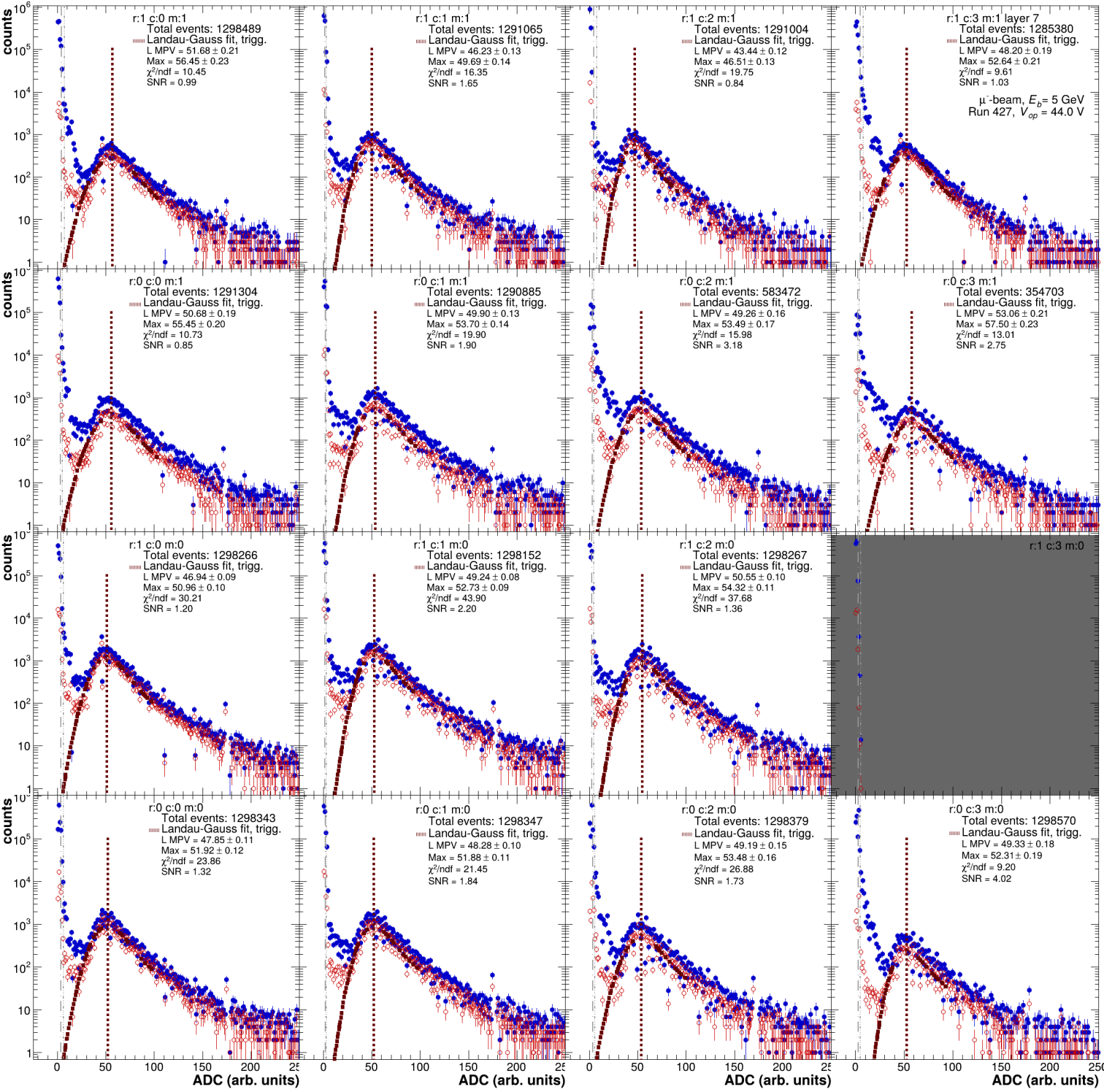

r:1 c:0 m:1
Total events: 654725
Landau-Gauss fit, trigg.
L MPV = 48.06 ± 0.17
Max = 52.77 ± 0.19
χ²/ndf = 24.07
SNR = 6.06
r:1 c:1 m:1
Total events: 1287111
Landau-Gauss fit, trigg.
L MPV = 49.08 ± 0.14
Max = 53.83 ± 0.15
χ²/ndf = 35.16
SNR = 5.84
r:1 c:2 m:1
Total events: 641876
Landau-Gauss fit, trigg.
L MPV = 51.67 ± 0.16
Max = 56.89 ± 0.18
χ²/ndf = 40.39
SNR = 10.36
r:1 c:3 m:1 layer 2
Total events: 188427
Landau-Gauss fit, trigg.
L MPV = 48.78 ± 0.16
Max = 53.72 ± 0.18
χ²/ndf = 43.80
SNR = 23.01
μ⁻-beam, E_b = 5 GeV
Run 427, V_op = 44.0 V
r:0 c:0 m:1
Total events: 1281791
Landau-Gauss fit, trigg.
L MPV = 46.46 ± 0.19
Max = 51.56 ± 0.21
χ²/ndf = 28.00
SNR = 2.91
r:0 c:1 m:1
Total events: 1288476
Landau-Gauss fit, trigg.
L MPV = 48.38 ± 0.16
Max = 53.63 ± 0.18
χ²/ndf = 36.24
SNR = 4.01
r:0 c:2 m:1
Total events: 1289427
Landau-Gauss fit, trigg.
L MPV = 46.23 ± 0.16
Max = 51.25 ± 0.17
χ²/ndf = 33.13
SNR = 3.57
r:0 c:3 m:1
Total events: 1290736
Landau-Gauss fit, trigg.
L MPV = 48.20 ± 0.23
Max = 53.49 ± 0.25
χ²/ndf = 19.94
SNR = 3.81
r:1 c:0 m:0
Total events: 1288546
Landau-Gauss fit, trigg.
L MPV = 46.80 ± 0.09
Max = 51.36 ± 0.10
χ²/ndf = 75.66
SNR = 4.78
r:1 c:1 m:0
Total events: 1287650
Landau-Gauss fit, trigg.
L MPV = 43.10 ± 0.07
Max = 47.06 ± 0.08
χ²/ndf = 65.20
SNR = 5.78
r:1 c:2 m:0
Total events: 1289937
Landau-Gauss fit, trigg.
L MPV = 48.72 ± 0.12
Max = 53.75 ± 0.13
χ²/ndf = 53.42
SNR = 19.31
r:1 c:3 m:0
Total events: 1294304
Landau-Gauss fit, trigg.
L MPV = 45.50 ± 0.28
Max = 51.73 ± 0.32
χ²/ndf = 22.70
SNR = 2.93
r:0 c:0 m:0
Total events: 1291788
Landau-Gauss fit, trigg.
L MPV = 47.81 ± 0.10
Max = 52.19 ± 0.11
χ²/ndf = 45.93
SNR = 5.08
r:0 c:1 m:0
Total events: 1291902
Landau-Gauss fit, trigg.
L MPV = 47.86 ± 0.11
Max = 52.51 ± 0.12
χ²/ndf = 52.50
SNR = 4.65
r:0 c:2 m:0
Total events: 1295313
Landau-Gauss fit, trigg.
L MPV = 47.22 ± 0.15
Max = 51.61 ± 0.16
χ²/ndf = 28.14
SNR = 4.45
r:0 c:3 m:0
Total events: 1296843
Landau-Gauss fit, trigg.
L MPV = 47.09 ± 0.23
Max = 51.78 ± 0.26
χ²/ndf = 14.71
SNR = 4.07
counts
ADC (arb. units)

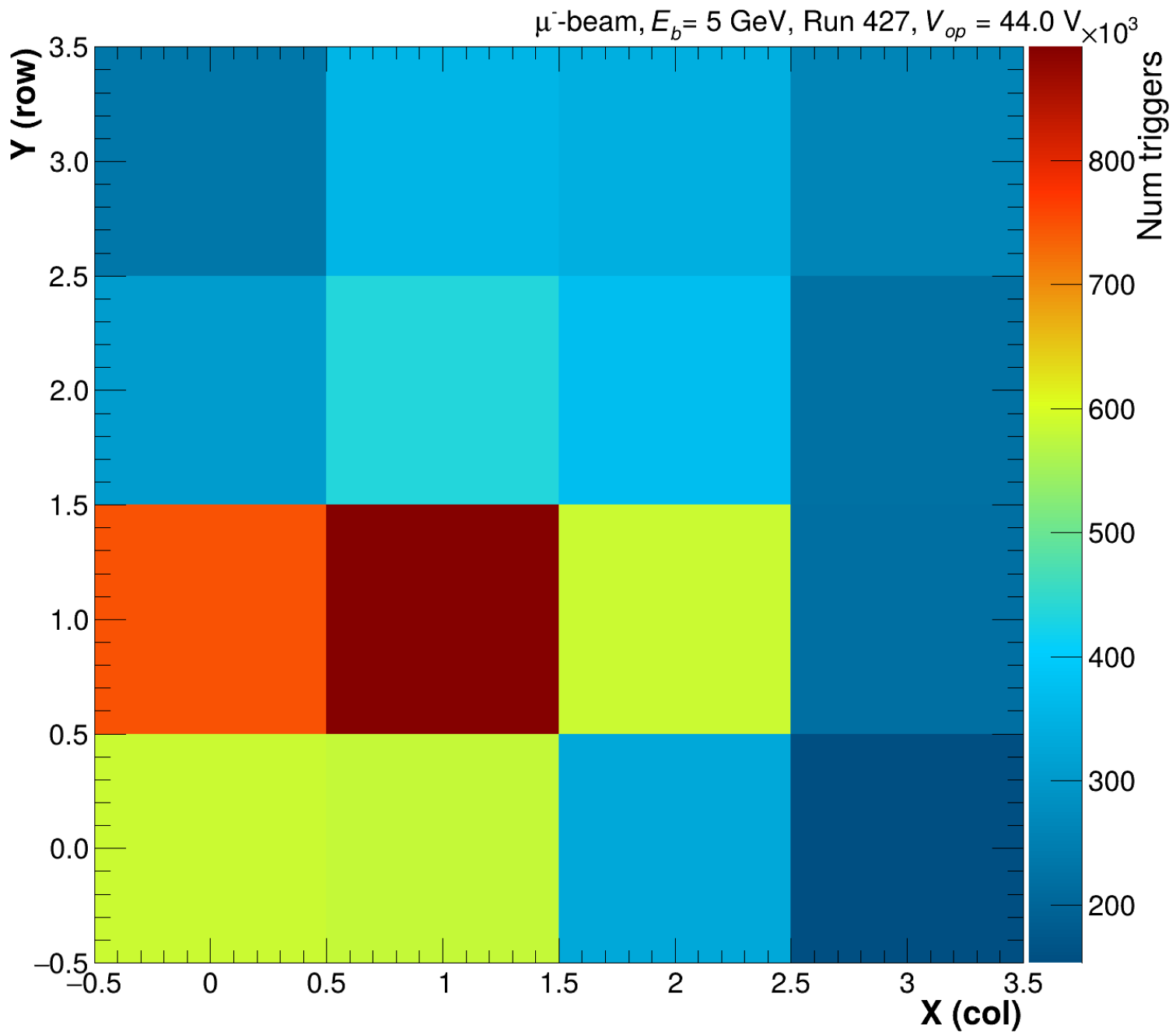

r:1 c:0 m:1
Total events: 1291100
Landau-Gauss fit, trigg.
L MPV = 49.47 ± 0.22
Max = 54.02 ± 0.24
χ²/ndf = 16.67
SNR = 1.41
r:1 c:1 m:1
Total events: 287435
Landau-Gauss fit, trigg.
L MPV = 53.60 ± 0.22
Max = 59.14 ± 0.25
χ²/ndf = 31.10
SNR = 3.83
r:1 c:2 m:1
Total events: 1291472
Landau-Gauss fit, trigg.
L MPV = 50.15 ± 0.19
Max = 54.86 ± 0.21
χ²/ndf = 11.86
SNR = 0.65
r:1 c:3 m:1 layer 3
Total events: 1298511
Landau-Gauss fit, trigg.
L MPV = 55.54 ± 0.25
Max = 60.76 ± 0.27
χ²/ndf = 12.63
SNR = 0.63
μ⁻-beam, E_b = 5 GeV
Run 427, V_op = 44.0 V
r:0 c:0 m:1
Total events: 1291054
Landau-Gauss fit, trigg.
L MPV = 47.74 ± 0.21
Max = 52.19 ± 0.23
χ²/ndf = 17.06
SNR = 1.49
r:0 c:1 m:1
Total events: 1284957
Landau-Gauss fit, trigg.
L MPV = 52.79 ± 0.23
Max = 58.25 ± 0.26
χ²/ndf = 20.03
SNR = 0.50
r:0 c:2 m:1
Total events: 582549
Landau-Gauss fit, trigg.
L MPV = 51.99 ± 0.25
Max = 57.66 ± 0.28
χ²/ndf = 19.08
SNR = 4.20
r:0 c:3 m:1
Total events: 1291175
Landau-Gauss fit, trigg.
L MPV = 54.87 ± 0.29
Max = 60.06 ± 0.31
χ²/ndf = 8.40
SNR = 0.66
r:1 c:0 m:0
Total events: 1298349
Landau-Gauss fit, trigg.
L MPV = 51.72 ± 0.13
Max = 56.66 ± 0.14
χ²/ndf = 71.28
SNR = 2.40
r:1 c:1 m:0
r:1 c:2 m:0
Total events: 1298328
Landau-Gauss fit, trigg.
L MPV = 52.81 ± 0.15
Max = 57.86 ± 0.16
χ²/ndf = 26.74
SNR = 0.56
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 1298501
Landau-Gauss fit, trigg.
L MPV = 48.56 ± 0.13
Max = 53.08 ± 0.14
χ²/ndf = 44.80
SNR = 2.02
r:0 c:1 m:0
Total events: 1298469
Landau-Gauss fit, trigg.
L MPV = 47.84 ± 0.14
Max = 52.46 ± 0.15
χ²/ndf = 22.57
SNR = 0.99
r:0 c:2 m:0
Total events: 1298491
Landau-Gauss fit, trigg.
L MPV = 52.76 ± 0.21
Max = 58.01 ± 0.23
χ²/ndf = 22.34
SNR = 1.58
r:0 c:3 m:0
Total events: 1298654
Landau-Gauss fit, trigg.
L MPV = 49.49 ± 0.27
Max = 54.18 ± 0.30
χ²/ndf = 6.03
SNR = 0.52
counts
ADC (arb. units)

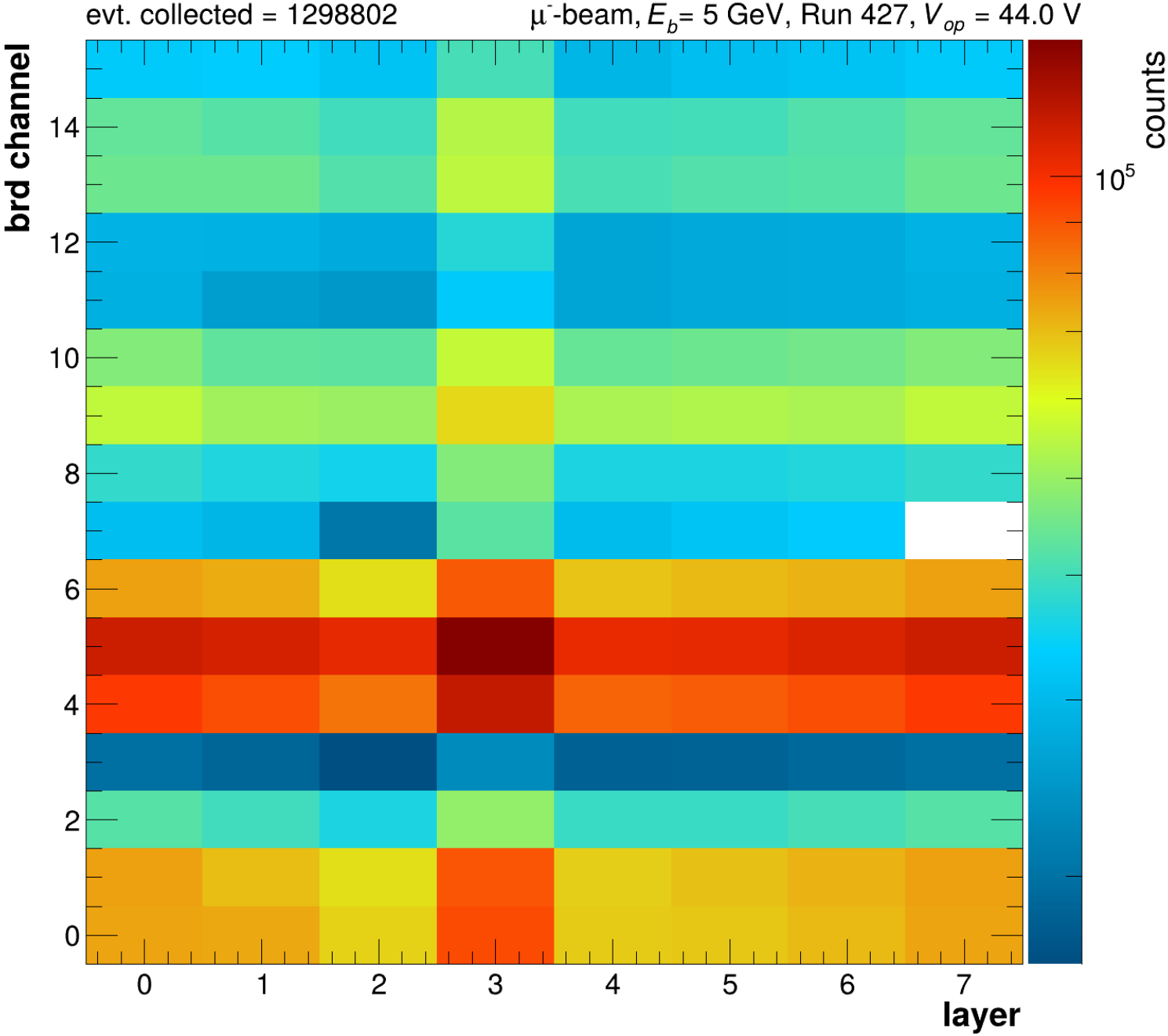

r:1 c:0 m:1
Total events: 1297504
Landau-Gauss fit, trigg.
L MPV = 50.75 ± 0.18
Max = 54.98 ± 0.19
χ²/ndf = 15.41
SNR = 2.74
r:1 c:1 m:1
Total events: 1292011
Landau-Gauss fit, trigg.
L MPV = 48.90 ± 0.14
Max = 53.05 ± 0.15
χ²/ndf = 23.07
SNR = 2.38
r:1 c:2 m:1
Total events: 1290070
Landau-Gauss fit, trigg.
L MPV = 47.44 ± 0.13
Max = 51.41 ± 0.14
χ²/ndf = 21.59
SNR = 2.20
r:1 c:3 m:1 layer 4
Total events: 1289503
Landau-Gauss fit, trigg.
L MPV = 55.96 ± 0.19
Max = 60.88 ± 0.21
χ²/ndf = 16.33
SNR = 2.28
μ⁻-beam, E_b = 5 GeV
Run 427, V_op = 44.0 V
r:0 c:0 m:1
Total events: 1289533
Landau-Gauss fit, trigg.
L MPV = 50.40 ± 0.17
Max = 54.78 ± 0.19
χ²/ndf = 21.61
SNR = 1.66
r:0 c:1 m:1
Total events: 1296418
Landau-Gauss fit, trigg.
L MPV = 48.63 ± 0.14
Max = 52.90 ± 0.15
χ²/ndf = 26.70
SNR = 1.84
r:0 c:2 m:1
Total events: 1288671
Landau-Gauss fit, trigg.
L MPV = 52.54 ± 0.18
Max = 57.29 ± 0.19
χ²/ndf = 21.75
SNR = 1.75
r:0 c:3 m:1
Total events: 717586
Landau-Gauss fit, trigg.
L MPV = 55.47 ± 0.17
Max = 59.29 ± 0.18
χ²/ndf = 8.27
SNR = 1.82
r:1 c:0 m:0
Total events: 1295028
Landau-Gauss fit, trigg.
L MPV = 48.83 ± 0.09
Max = 53.01 ± 0.10
χ²/ndf = 58.40
SNR = 2.70
r:1 c:1 m:0
Total events: 1293398
Landau-Gauss fit, trigg.
L MPV = 48.86 ± 0.08
Max = 53.13 ± 0.09
χ²/ndf = 54.21
SNR = 2.46
r:1 c:2 m:0
Total events: 1295327
Landau-Gauss fit, trigg.
L MPV = 48.97 ± 0.10
Max = 53.16 ± 0.11
χ²/ndf = 37.35
SNR = 2.25
r:1 c:3 m:0
Total events: 1296798
Landau-Gauss fit, trigg.
L MPV = 52.84 ± 0.19
Max = 57.87 ± 0.21
χ²/ndf = 18.54
SNR = 1.76
r:0 c:0 m:0
Total events: 1296012
Landau-Gauss fit, trigg.
L MPV = 49.64 ± 0.08
Max = 52.59 ± 0.08
χ²/ndf = 31.18
SNR = 2.49
r:0 c:1 m:0
Total events: 1295346
Landau-Gauss fit, trigg.
L MPV = 55.41 ± 0.13
Max = 60.26 ± 0.15
χ²/ndf = 41.15
SNR = 2.26
r:0 c:2 m:0
Total events: 1297069
Landau-Gauss fit, trigg.
L MPV = 49.88 ± 0.13
Max = 53.95 ± 0.14
χ²/ndf = 22.52
SNR = 2.60
r:0 c:3 m:0
Total events: 1297936
Landau-Gauss fit, trigg.
L MPV = 49.45 ± 0.17
Max = 53.07 ± 0.18
χ²/ndf = 8.03
SNR = 2.56
counts
ADC (arb. units)

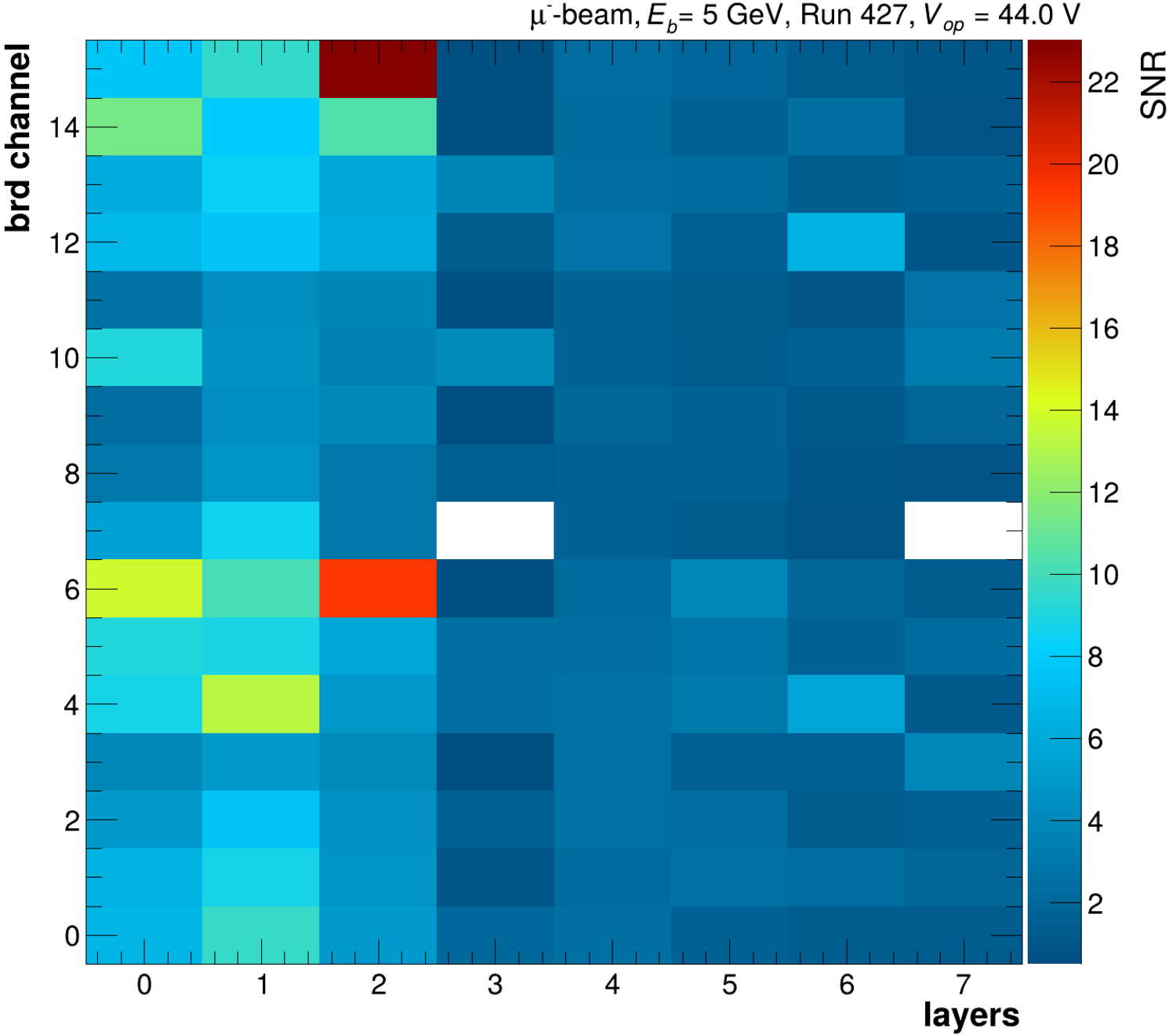

r:1 c:0 m:1
Total events: 1290801
Landau-Gauss fit, trigg.
L MPV = 54.76 ± 0.16
Max = 58.51 ± 0.17
χ²/ndf = 16.27
SNR = 1.56
r:1 c:1 m:1
Total events: 1290567
Landau-Gauss fit, trigg.
L MPV = 53.30 ± 0.12
Max = 56.93 ± 0.13
χ²/ndf = 18.05
SNR = 2.25
r:1 c:2 m:1
Total events: 584156
Landau-Gauss fit, trigg.
L MPV = 61.49 ± 0.16
Max = 66.05 ± 0.17
χ²/ndf = 19.75
SNR = 1.61
r:1 c:3 m:1 layer 5
Total events: 1290389
Landau-Gauss fit, trigg.
L MPV = 55.92 ± 0.19
Max = 60.52 ± 0.20
χ²/ndf = 15.42
SNR = 1.92
μ⁻-beam, E_b = 5 GeV
Run 427, V_op = 44.0 V
r:0 c:0 m:1
Total events: 1297756
Landau-Gauss fit, trigg.
L MPV = 57.28 ± 0.16
Max = 61.40 ± 0.17
χ²/ndf = 14.09
SNR = 1.72
r:0 c:1 m:1
Total events: 1290132
Landau-Gauss fit, trigg.
L MPV = 52.93 ± 0.12
Max = 56.53 ± 0.13
χ²/ndf = 24.58
SNR = 1.79
r:0 c:2 m:1
Total events: 1285080
Landau-Gauss fit, trigg.
L MPV = 58.08 ± 0.20
Max = 62.76 ± 0.21
χ²/ndf = 20.81
SNR = 1.29
r:0 c:3 m:1
Total events: 1291016
Landau-Gauss fit, trigg.
L MPV = 55.97 ± 0.25
Max = 60.24 ± 0.27
χ²/ndf = 16.21
SNR = 1.44
r:1 c:0 m:0
Total events: 1296753
Landau-Gauss fit, trigg.
L MPV = 57.84 ± 0.09
Max = 61.94 ± 0.10
χ²/ndf = 51.79
SNR = 3.03
r:1 c:1 m:0
Total events: 1295974
Landau-Gauss fit, trigg.
L MPV = 60.28 ± 0.09
Max = 64.64 ± 0.10
χ²/ndf = 48.61
SNR = 2.84
r:1 c:2 m:0
Total events: 1297203
Landau-Gauss fit, trigg.
L MPV = 52.04 ± 0.10
Max = 55.79 ± 0.11
χ²/ndf = 29.47
SNR = 3.92
r:1 c:3 m:0
Total events: 1297901
Landau-Gauss fit, trigg.
L MPV = 56.44 ± 0.21
Max = 61.30 ± 0.22
χ²/ndf = 13.07
SNR = 1.22
r:0 c:0 m:0
Total events: 1297171
Landau-Gauss fit, trigg.
L MPV = 60.38 ± 0.13
Max = 64.93 ± 0.13
χ²/ndf = 34.54
SNR = 1.70
r:0 c:1 m:0
Total events: 1297381
Landau-Gauss fit, trigg.
L MPV = 55.49 ± 0.10
Max = 59.31 ± 0.11
χ²/ndf = 47.20
SNR = 2.60
r:0 c:2 m:0
Total events: 1297756
Landau-Gauss fit, trigg.
L MPV = 60.67 ± 0.17
Max = 65.24 ± 0.18
χ²/ndf = 18.40
SNR = 2.37
r:0 c:3 m:0
Total events: 1298288
Landau-Gauss fit, trigg.
L MPV = 57.21 ± 0.23
Max = 61.52 ± 0.25
χ²/ndf = 9.01
SNR = 1.48
counts
ADC (arb. units)

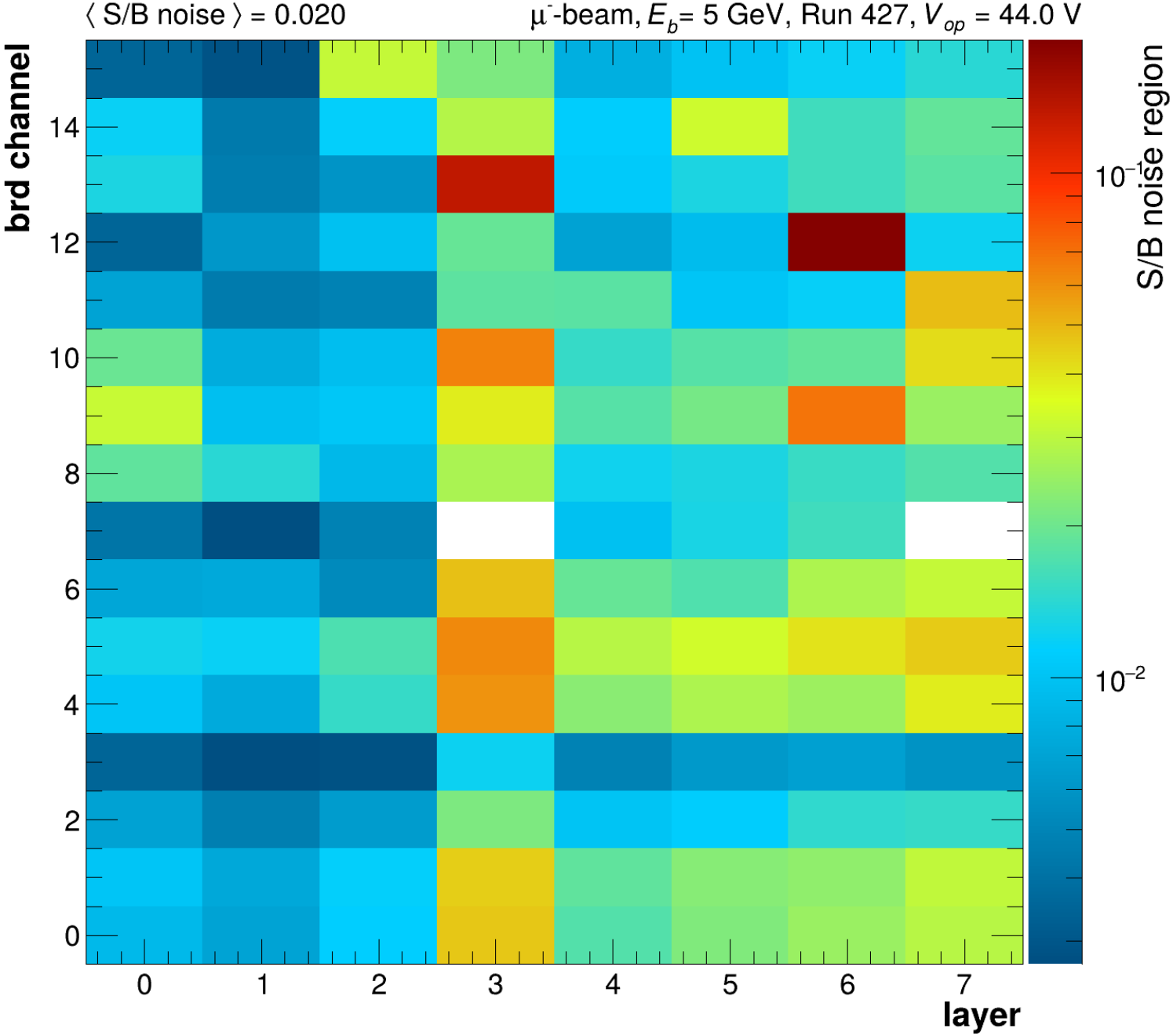

r:1 c:0 m:1
Total events: 103105
Landau-Gauss fit, trigg.
L MPV = 68.14 ± 0.26
Max = 73.35 ± 0.28
χ²/ndf = 19.66
SNR = 6.50
r:1 c:1 m:1
Total events: 1290870
Landau-Gauss fit, trigg.
L MPV = 67.77 ± 0.19
Max = 72.66 ± 0.20
χ²/ndf = 15.98
SNR = 1.42
r:1 c:2 m:1
Total events: 1290779
Landau-Gauss fit, trigg.
L MPV = 62.13 ± 0.17
Max = 66.64 ± 0.18
χ²/ndf = 25.80
SNR = 2.49
r:1 c:3 m:1 layer 6
Total events: 1291209
Landau-Gauss fit, trigg.
L MPV = 68.88 ± 0.23
Max = 73.99 ± 0.25
χ²/ndf = 11.31
SNR = 1.33
μ⁻-beam, E_b = 5 GeV
Run 427, V_op = 44.0 V
r:0 c:0 m:1
Total events: 1298130
Landau-Gauss fit, trigg.
L MPV = 64.48 ± 0.23
Max = 69.42 ± 0.24
χ²/ndf = 15.89
SNR = 1.07
r:0 c:1 m:1
Total events: 477730
Landau-Gauss fit, trigg.
L MPV = 71.82 ± 0.23
Max = 77.42 ± 0.25
χ²/ndf = 31.39
SNR = 1.18
r:0 c:2 m:1
Total events: 1290680
Landau-Gauss fit, trigg.
L MPV = 64.93 ± 0.19
Max = 69.79 ± 0.21
χ²/ndf = 18.86
SNR = 1.80
r:0 c:3 m:1
Total events: 1290971
Landau-Gauss fit, trigg.
L MPV = 68.69 ± 0.28
Max = 73.86 ± 0.30
χ²/ndf = 10.47
SNR = 0.94
r:1 c:0 m:0
Total events: 1297720
Landau-Gauss fit, trigg.
L MPV = 61.07 ± 0.11
Max = 65.61 ± 0.12
χ²/ndf = 65.18
SNR = 5.62
r:1 c:1 m:0
Total events: 1297426
Landau-Gauss fit, trigg.
L MPV = 64.91 ± 0.11
Max = 69.66 ± 0.11
χ²/ndf = 44.51
SNR = 1.79
r:1 c:2 m:0
Total events: 1297892
Landau-Gauss fit, trigg.
L MPV = 63.08 ± 0.12
Max = 67.65 ± 0.13
χ²/ndf = 48.83
SNR = 1.85
r:1 c:3 m:0
Total events: 1298330
Landau-Gauss fit, trigg.
L MPV = 61.14 ± 0.29
Max = 66.30 ± 0.31
χ²/ndf = 9.97
SNR = 0.77
r:0 c:0 m:0
Total events: 1298022
Landau-Gauss fit, trigg.
L MPV = 66.01 ± 0.15
Max = 71.06 ± 0.16
χ²/ndf = 28.50
SNR = 1.45
r:0 c:1 m:0
Total events: 1297998
Landau-Gauss fit, trigg.
L MPV = 59.77 ± 0.13
Max = 64.32 ± 0.14
χ²/ndf = 53.76
SNR = 2.40
r:0 c:2 m:0
Total events: 1298284
Landau-Gauss fit, trigg.
L MPV = 62.64 ± 0.19
Max = 67.31 ± 0.20
χ²/ndf = 25.77
SNR = 1.32
r:0 c:3 m:0
Total events: 1298521
Landau-Gauss fit, trigg.
L MPV = 60.16 ± 0.23
Max = 64.51 ± 0.25
χ²/ndf = 7.78
SNR = 1.60
counts
ADC (arb. units)

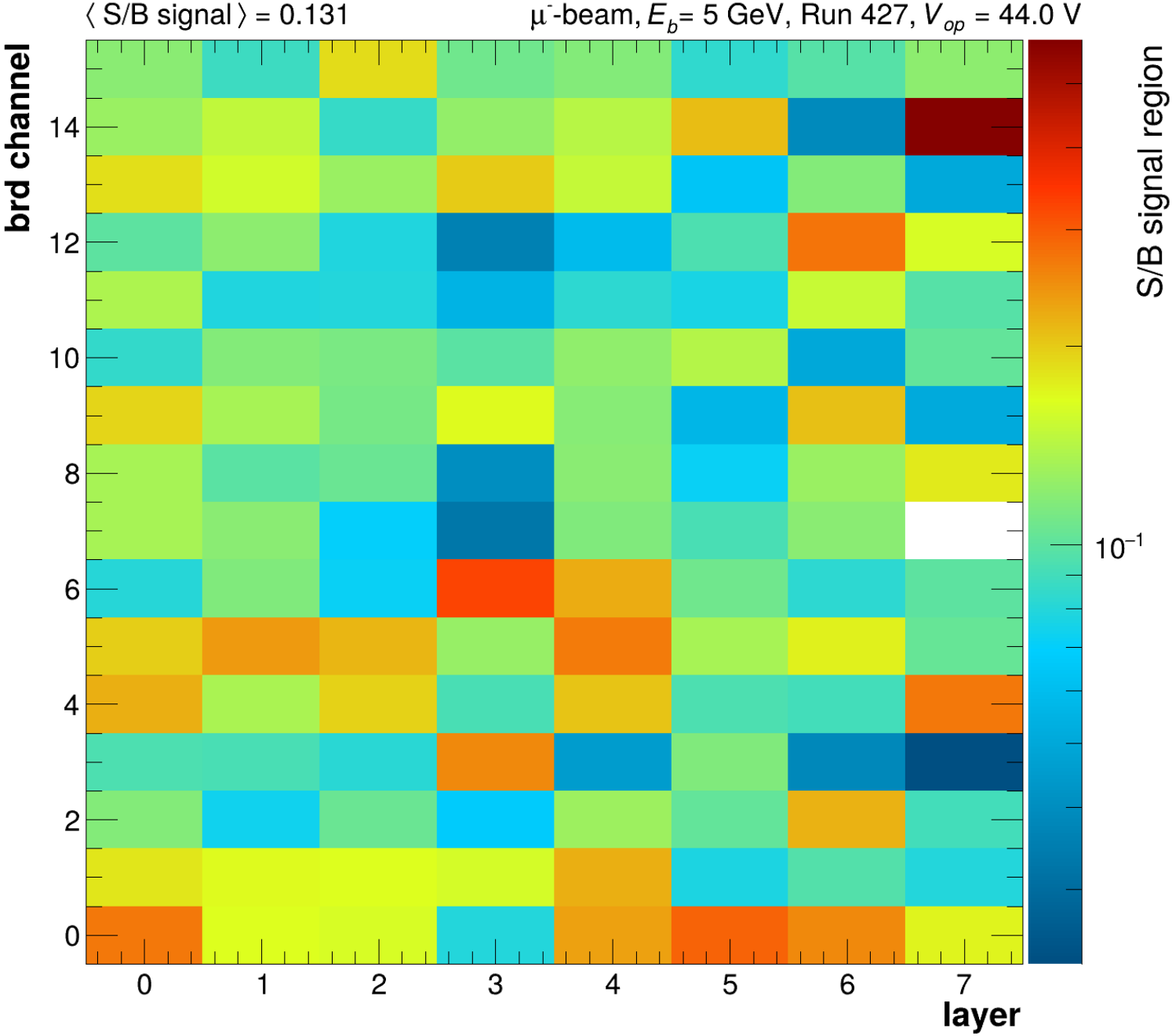

r:1 c:0 m:1
Total events: 1298489
Landau-Gauss fit, trigg.
L MPV = 51.68 ± 0.21
Max = 56.45 ± 0.23
χ²/ndf = 10.45
SNR = 0.99
r:1 c:1 m:1
Total events: 1291065
Landau-Gauss fit, trigg.
L MPV = 46.23 ± 0.13
Max = 49.69 ± 0.14
χ²/ndf = 16.35
SNR = 1.65
r:1 c:2 m:1
Total events: 1291004
Landau-Gauss fit, trigg.
L MPV = 43.44 ± 0.12
Max = 46.51 ± 0.13
χ²/ndf = 19.75
SNR = 0.84
r:1 c:3 m:1 layer 7
Total events: 1285380
Landau-Gauss fit, trigg.
L MPV = 48.20 ± 0.19
Max = 52.64 ± 0.21
χ²/ndf = 9.61
SNR = 1.03
μ⁻-beam, E_b = 5 GeV
Run 427, V_op = 44.0 V
r:0 c:0 m:1
Total events: 1291304
Landau-Gauss fit, trigg.
L MPV = 50.68 ± 0.19
Max = 55.45 ± 0.20
χ²/ndf = 10.73
SNR = 0.85
r:0 c:1 m:1
Total events: 1290885
Landau-Gauss fit, trigg.
L MPV = 49.90 ± 0.13
Max = 53.70 ± 0.14
χ²/ndf = 19.90
SNR = 1.90
r:0 c:2 m:1
Total events: 583472
Landau-Gauss fit, trigg.
L MPV = 49.26 ± 0.16
Max = 53.49 ± 0.17
χ²/ndf = 15.98
SNR = 3.18
r:0 c:3 m:1
Total events: 354703
Landau-Gauss fit, trigg.
L MPV = 53.06 ± 0.21
Max = 57.50 ± 0.23
χ²/ndf = 13.01
SNR = 2.75
r:1 c:0 m:0
Total events: 1298266
Landau-Gauss fit, trigg.
L MPV = 46.94 ± 0.09
Max = 50.96 ± 0.10
χ²/ndf = 30.21
SNR = 1.20
r:1 c:1 m:0
Total events: 1298152
Landau-Gauss fit, trigg.
L MPV = 49.24 ± 0.08
Max = 52.73 ± 0.09
χ²/ndf = 43.90
SNR = 2.20
r:1 c:2 m:0
Total events: 1298267
Landau-Gauss fit, trigg.
L MPV = 50.55 ± 0.10
Max = 54.32 ± 0.11
χ²/ndf = 37.68
SNR = 1.36
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 1298343
Landau-Gauss fit, trigg.
L MPV = 47.85 ± 0.11
Max = 51.92 ± 0.12
χ²/ndf = 23.86
SNR = 1.32
r:0 c:1 m:0
Total events: 1298347
Landau-Gauss fit, trigg.
L MPV = 48.28 ± 0.10
Max = 51.88 ± 0.11
χ²/ndf = 21.45
SNR = 1.84
r:0 c:2 m:0
Total events: 1298379
Landau-Gauss fit, trigg.
L MPV = 49.19 ± 0.15
Max = 53.48 ± 0.16
χ²/ndf = 26.88
SNR = 1.73
r:0 c:3 m:0
Total events: 1298570
Landau-Gauss fit, trigg.
L MPV = 49.33 ± 0.18
Max = 52.31 ± 0.19
χ²/ndf = 9.20
SNR = 4.02
counts
ADC (arb. units)

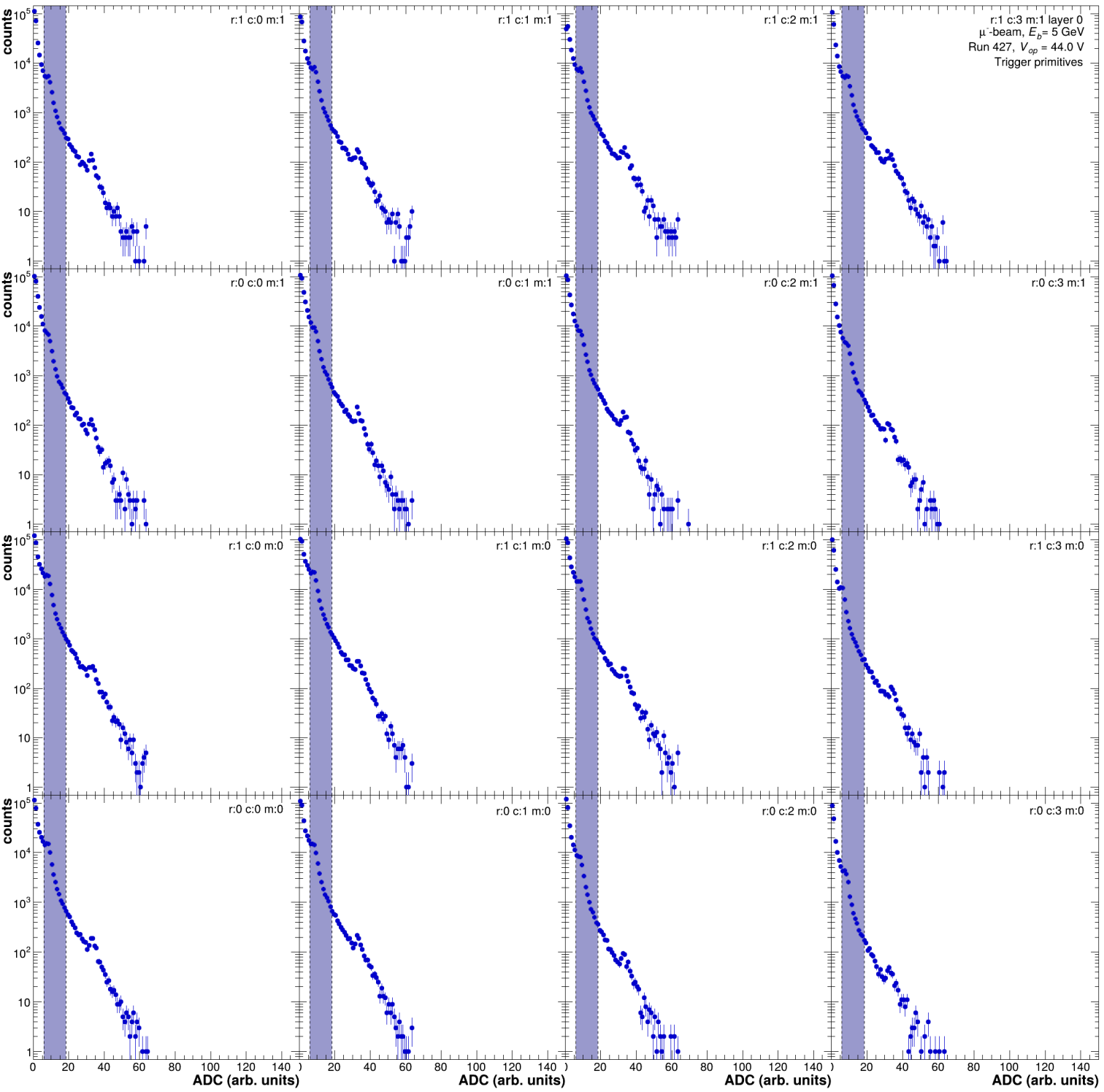

μ⁻-beam, E_b= 5 GeV, Run 427, V_{op} = 44.0 V
Y (row)
X (col)
Num triggers
×10³
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−0.5
0
0.5
1
1.5
2
2.5
3
3.5
800
700
600
500
400
300
200

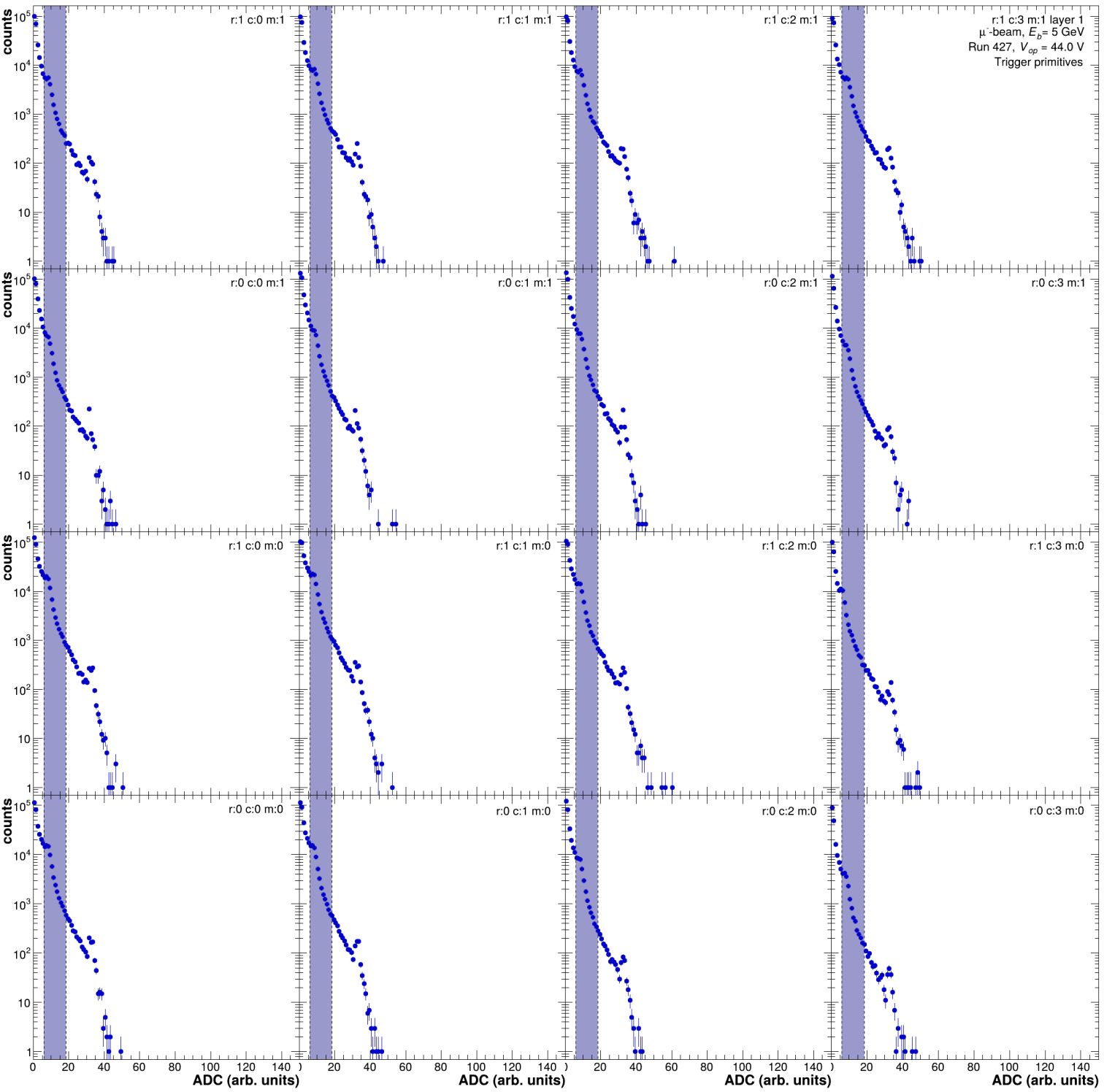

evt. collected = 1298802
μ⁻-beam, E_b= 5 GeV, Run 427, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
counts
10^5

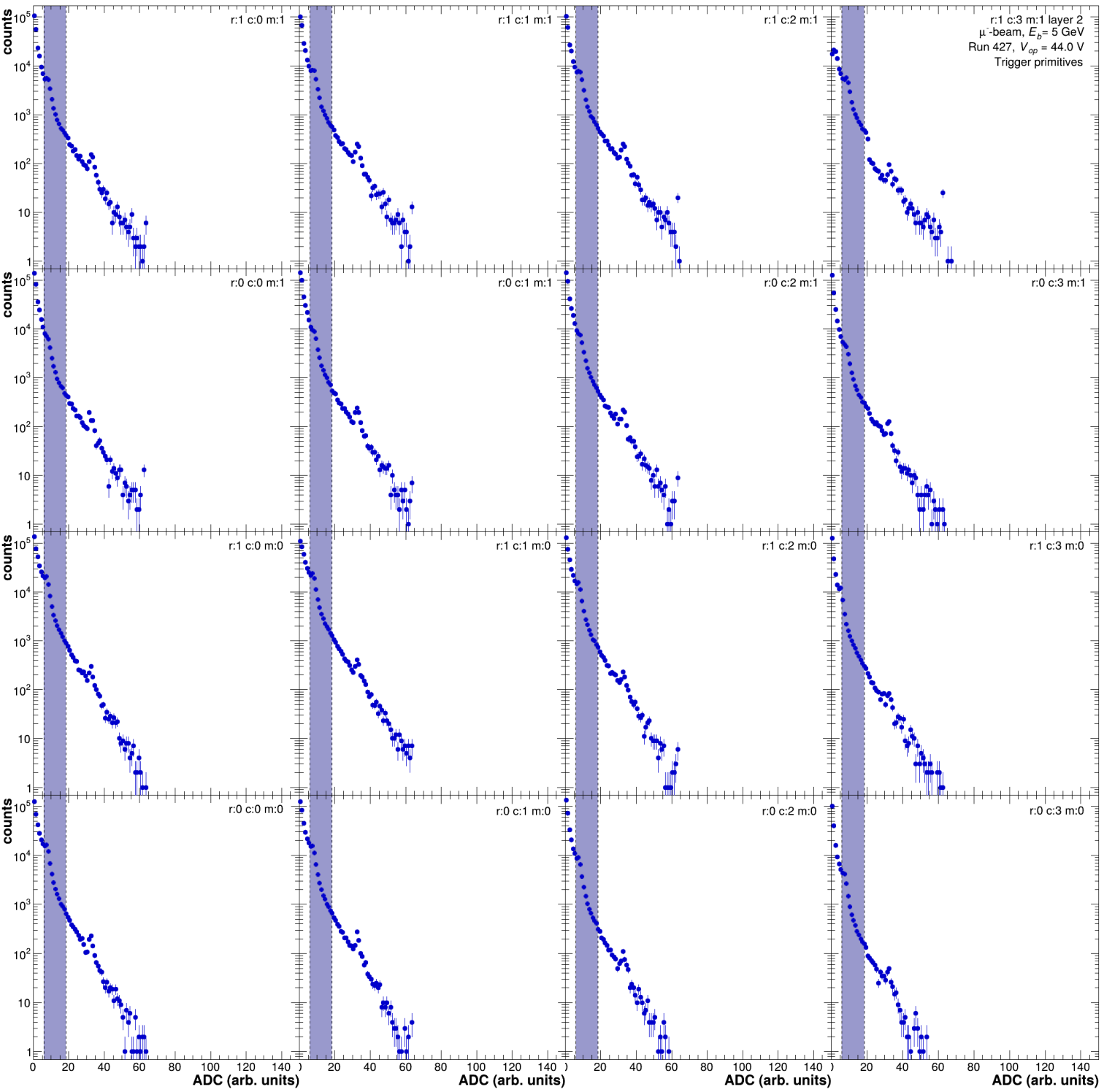

μ⁻-beam, E_b= 5 GeV, Run 427, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layers
SNR
2
4
6
8
10
12
14
16
18
20
22

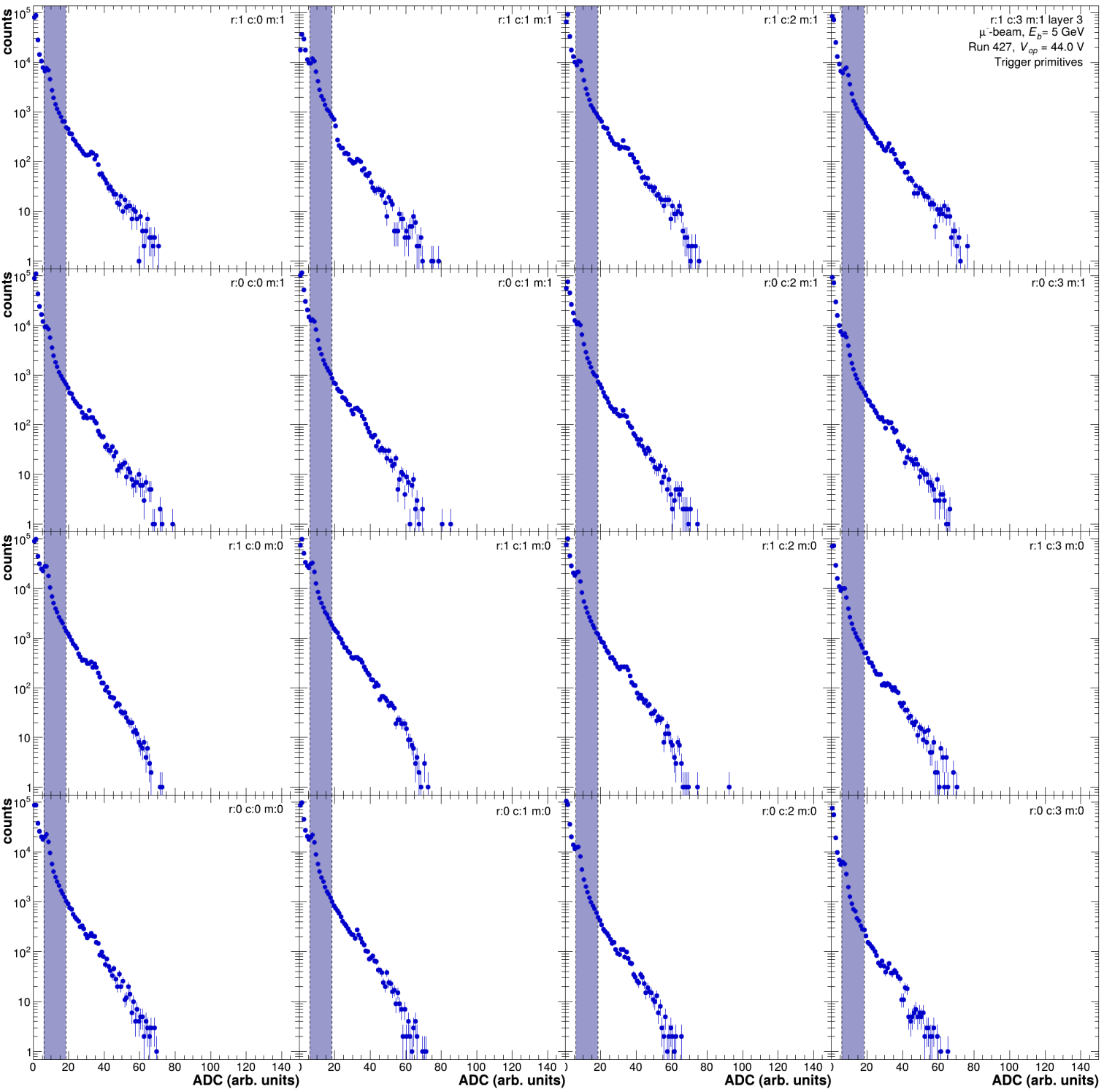

⟨ S/B noise ⟩ = 0.020
μ⁻-beam, E_b= 5 GeV, Run 427, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
S/B noise region
10^{-1}
10^{-2}

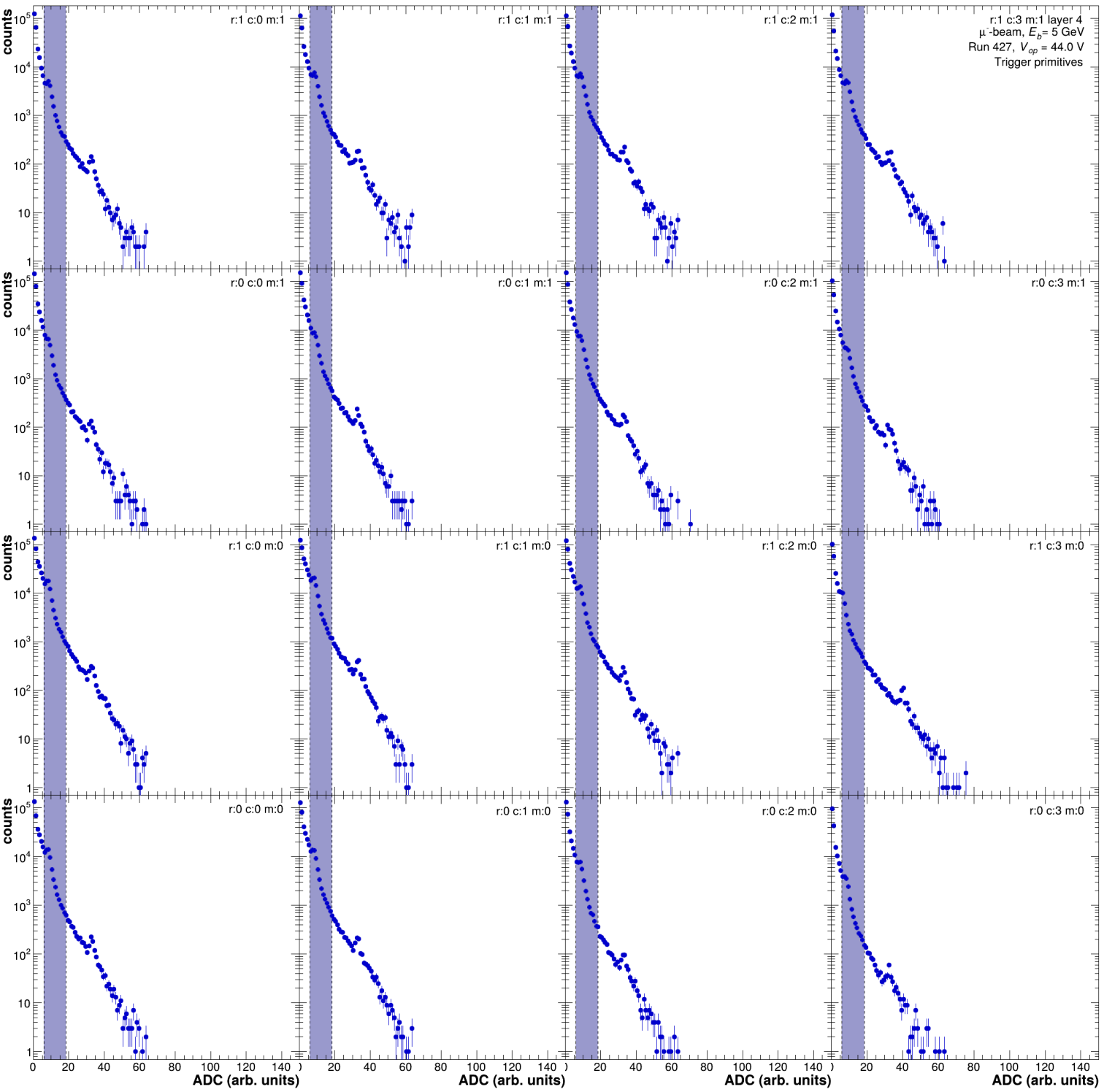

⟨ S/B signal ⟩ = 0.131
μ⁻-beam, E_b= 5 GeV, Run 427, V_{op} = 44.0 V
brd channel
layer
S/B signal region
10^{-1}
0
1
2
3
4
5
6
7
0
2
4
6
8
10
12
14

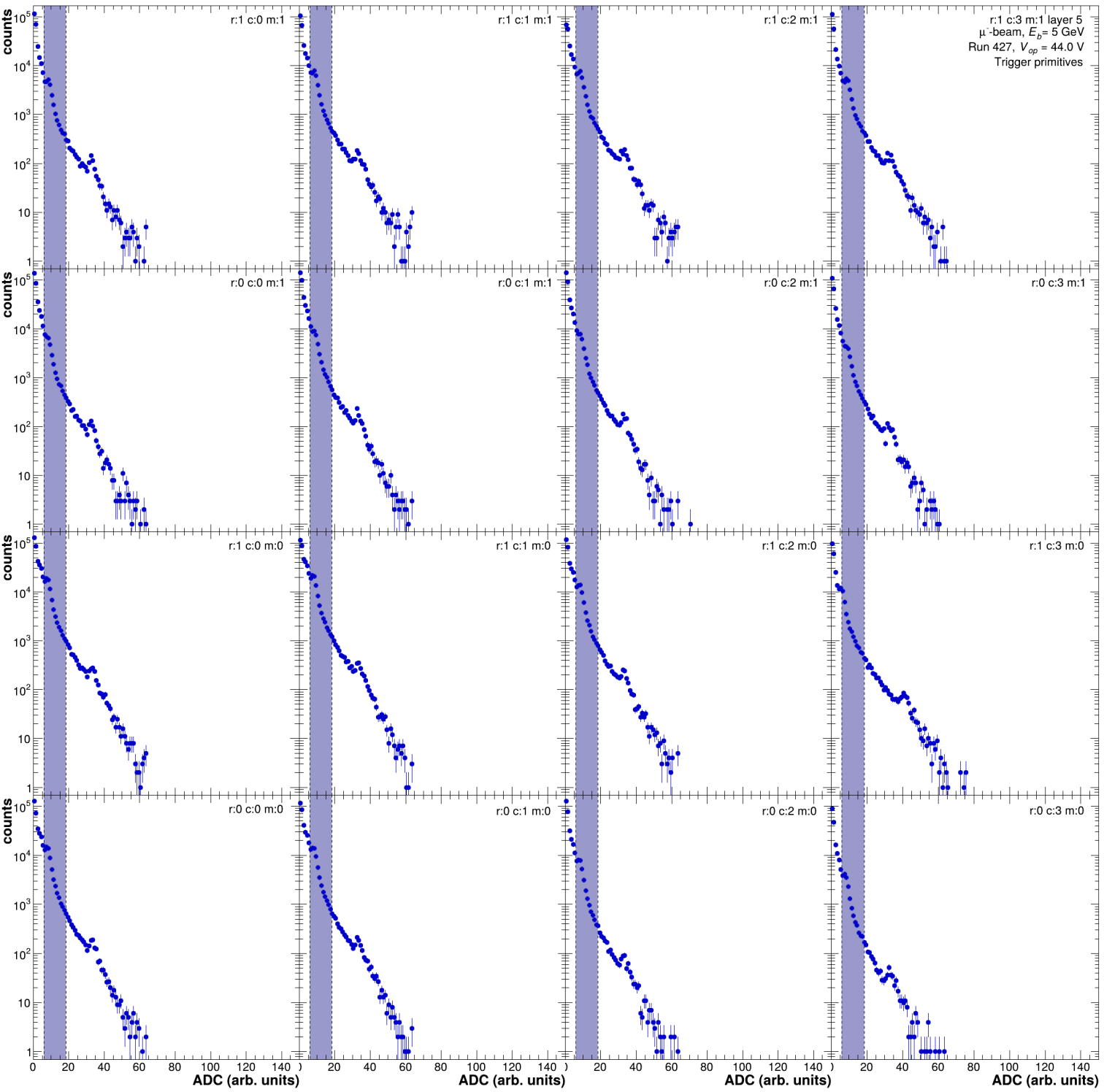

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 0
μ⁻-beam, E_b= 5 GeV
Run 427, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

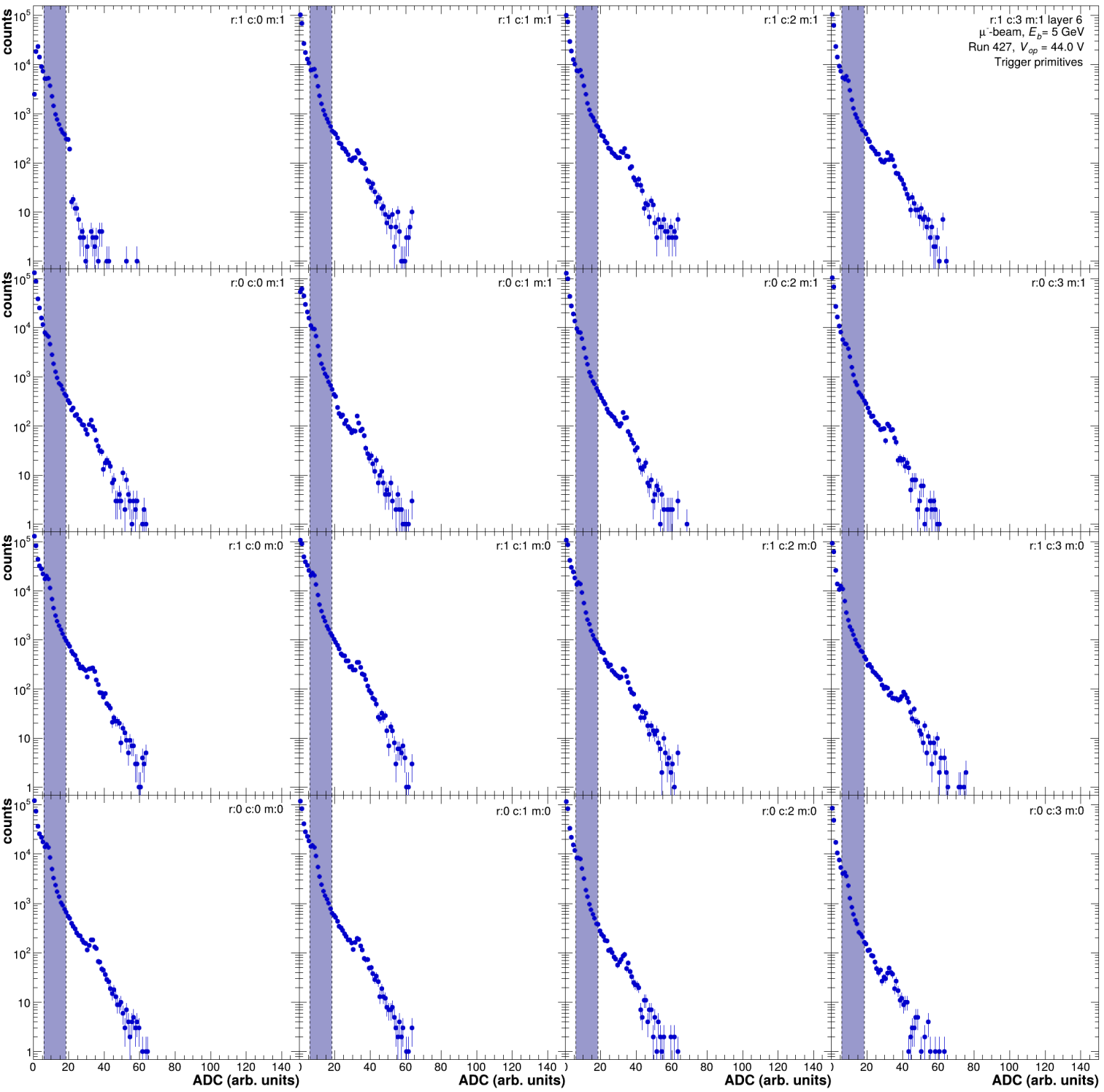

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 1
μ⁻-beam, E_b= 5 GeV
Run 427, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

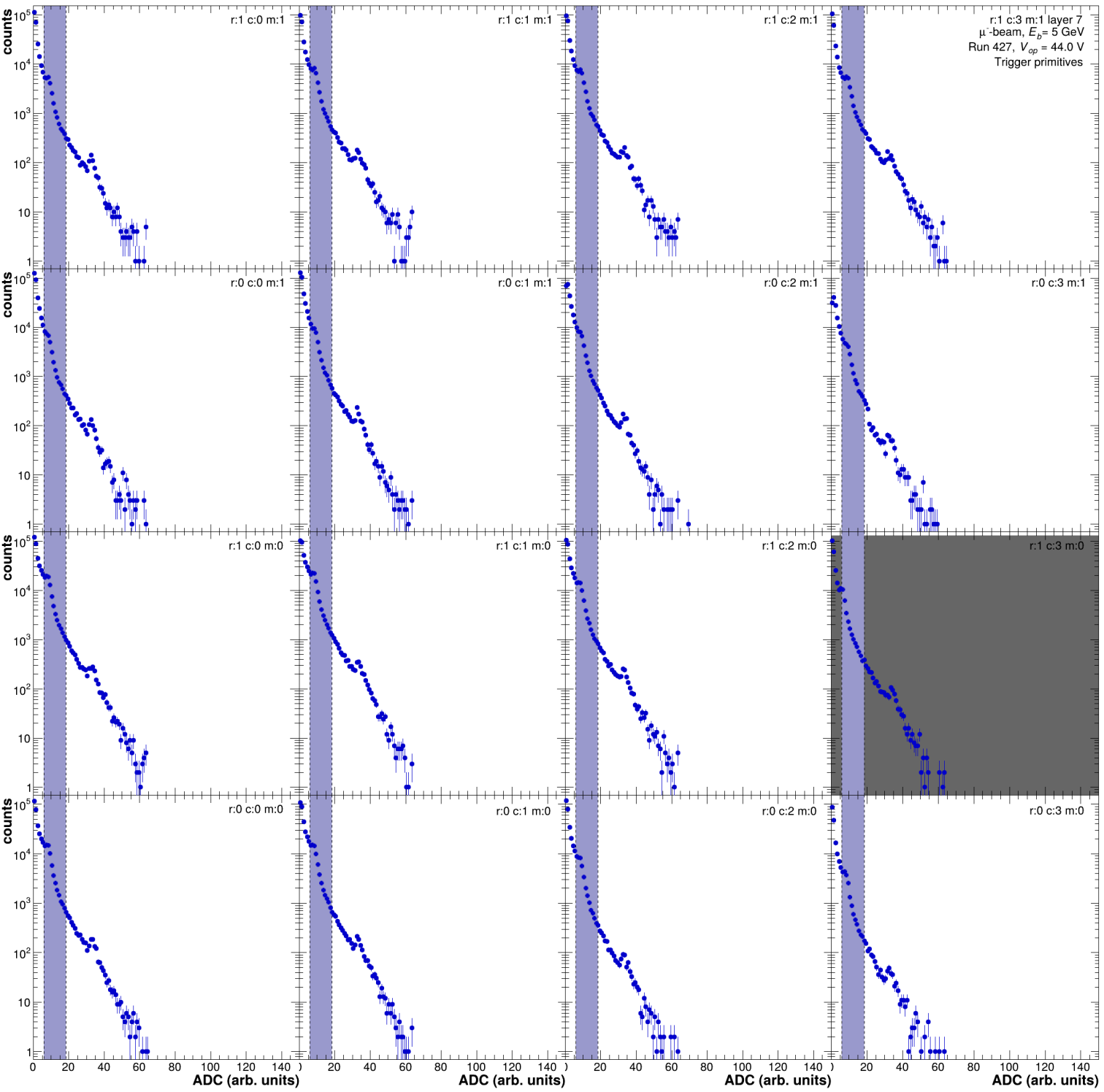

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 2
μ⁻-beam, E_b= 5 GeV
Run 427, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 3
μ⁻-beam, E_b= 5 GeV
Run 427, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 4
μ⁻-beam, E_b= 5 GeV
Run 427, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 5
μ⁻-beam, E_b= 5 GeV
Run 427, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 6
μ⁻-beam, E_b= 5 GeV
Run 427, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 7
μ⁻-beam, E_b= 5 GeV
Run 427, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)